

OPERATION

NOTE: BEFORE WORKING, CLEANING OR MAINTAINING THE EQUIPMENT, IT IS HIGHLY IMPORTANT TO LOCK OUT ALL OF THE ENERGY SOURCES ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PRE STARTUP CHECKLIST

Electrical Checklist (follow checklist every time the Chip Fryer is to be operated)

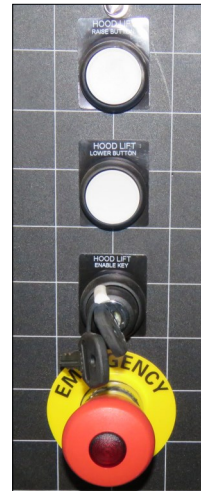
- ☐ Ensure that all electrical safety interlocks on electrical power panels, doors, guards, and covers are engaged.
- ☐ Assure that all electrical buttons are in their off positions (including those on remote locations such as Emergency Stop buttons).
- ☐ Ensure main circuit breaker is in the ON position.
- ☐ Ansul fire suppression system should be installed to local safety regulations and fire code **Prior to Operating Fryer.**
- ☐ Follow Pre-Start Up Checklist for all other machines as described in their individual Lawrence Equipment technical manual. Do not attempt to run this machine without pre-checking all machinery.

Mechanical Checklist (follow checklist every time the Chip Fryer is to be operated)

- ☐ Before starting the Chip Fryer it should be inspected for any abnormal wear and tension.
- ☐ Verify that all sprockets (and keys) are:
 - aligned properly.
 - tight and secured in their respective positions.
 - tracking correctly with their respective chains.
- ☐ Verify that all conveyor belts, drive chains, sprockets and pulleys are free of any debris, tools or any other obstructions which may impair the proper operation of the machine.
- ☐ Verify that all safety and access panels, guards and covers are:
 - in their proper place.
 - closed and/or secured.
- ☐ Establish that no one is working on the machine.
- ☐ Establish that hands, loose clothing or any other objects are out of the machine's proximity.
- ☐ Follow Pre-Start Up Checklist for all other machines as described in their individual Lawrence Equipment technical manual. Do not attempt to run this machine without pre-checking all machinery.
- ☐ Assure that all valves are in the correct position. Refer to valve position chart.
- ☐ Assure that hi-limit alarm temperature unit is set to about 10 degrees above normal operating temperature.
- ☐ Assure that the cooking tank is clean, dry and free of dirt, dust and moisture.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER CONTROL PANEL



1. **Main Disconnect Switch**- This switch disconnects the main electrical power to the Chip Fryer, when the switch is in the ON position the main control panel will be electrically energized. Always turn the disconnect switch to the off position, and disconnect the main electrical power to the machine before working on and or cleaning the machine.
2. **Safety System Reset**: Pressing this button energizes the safety relay and allows power flow through the control panel. There are three different settings for this Safety Reset Button. A Blue light illuminated means an alarm has faulted and system must be cleared. A blinking light means alarm has been cleared, showing system is ready to be reset. Safety System Reset indicator light off means the control system power is present and system is ready to run.
3. **Fryer Stop**: Activating this button will stop the machine.
4. **Hood Lift Raise**: Pressing this button allows the user to raise the Chip Fryer hood. Hood lift key must be enable for this button to function.
5. **Hood Lift Lower**: Pressing this button allows the user to lower the Chip Fryer hood. Hood lift key must be enable for this button to function.
6. **Hood Lift Enable Key**: Turning the selector key enables the "hood lift" to "Raise" or "Lower" the Fryer hood when either button is being activated.
7. **Emergency Stops and Remote Location E-Stops**: Pushing these buttons De-Energizes the control system. If the E-Stop button (s) are pushed they will be illuminated red to indicate they are active. If you wish to restart the system locate E-stop active on the control panel by accessing the Alarm Screen then walk to location to pull E-Stop button (s) out Panic Bars should be back into place. Once all E-stops are deactivated the Safety System Energize button should blink blue showing system is ready to



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER CONTROL PANEL (continued)

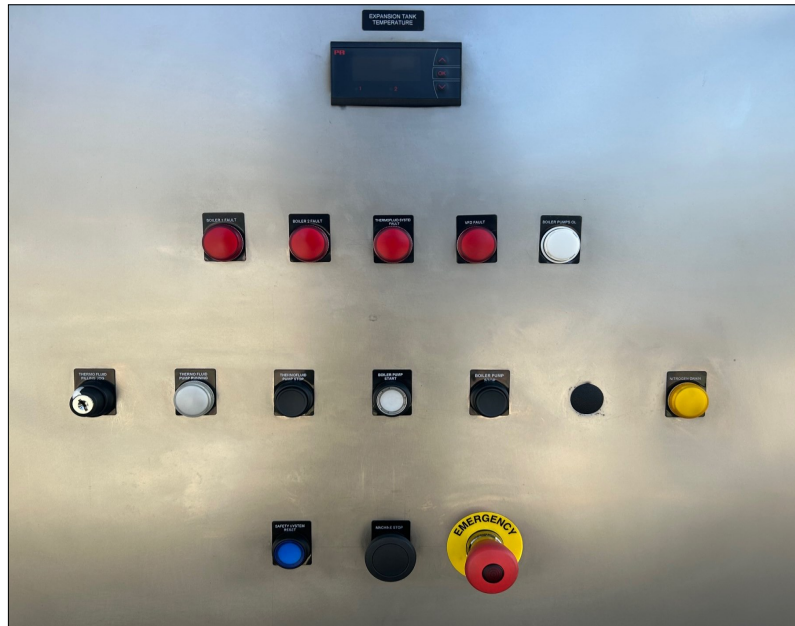
be reset.

ALWAYS remember:

1. Prior to running the Chip Fryer, make sure that all safety features are in place and that none have been bypassed. This will not only void any warranties but ensure operator safety.
2. Save all literature and instructions for future use.
3. **DO NOT** remove any placards on this machine.
4. If you are not familiar with the operation of this machine, DO NOT attempt to operate it until you have received proper training.
5. **DO NOT** spray water on any of the electrical components, as this can cause electrical shorting and injury to the operator.
6. This is a precision machine; follow all instructions on the care and use of this machine. Make sure that you are familiar with the functions of the machine, as this will extend the life of the machine.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

THERMOFLUID SYSTEM CONTROL PANEL



1. **Safety System Reset:** When momentarily selected, this energizes the Safety System when all safety requirements are made. The lens will blink blue when ready to reset. If lens is not blinking check all emergency stop buttons, they should all be disengaged (pulled out) The e stop will be illuminated if engaged.
2. **Machine Stop:** When momentarily selected, this disables the local machine safety system. In this case, all the boilers, the boiler pumps and the ThermoFluid pump will be de energized. This does NOT disable all other machines in the line.
3. **Emergency Stop:** When depressed and engaged (pushed in) this de energizes ALL of the machinery on the line. This will shut off all pumps, motors and control systems on the fryer, support module and the boilers in the event of a catastrophic failure or emergency situation.
4. **Expansion Tank Temperature:** This digital display is the current ThermoFluid temperature in the Expansion Tank above the boiler module. This temperature should not rise above 190 degrees Fahrenheit. This control is set to disable the boilers if the temperature rises above 195 degrees Fahrenheit. DO NOT restart the boilers until the condition that caused the high temperature is rectified.
5. **Boiler 1 Fault:** This indicator will illuminate when a fault occurs to shut down the boiler 1. This includes a flame controller fault, over temperature, pump shutdown or pressure differential switch indication.

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

THERMOFLUID SYSTEM CONTROL PANEL

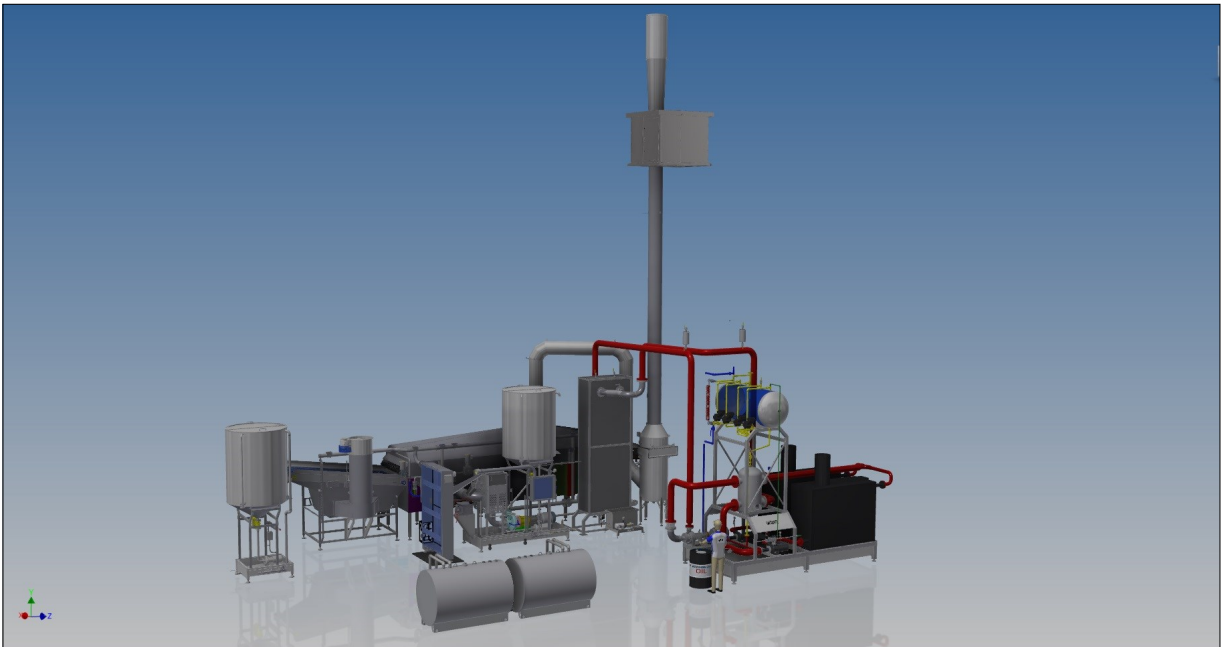
6. **Boiler 2 Fault:** This indicator will illuminate when a fault occurs to shut down the boiler 2. This includes a flame controller fault, over temperature, pump shutdown or pressure differential switch indication.
7. **Thermofluid System Fault:** When illuminated Indicates there is a problem with the Thermofluid system. Investigate the problem before restarting.
8. **VFD Fault:** When illuminated indicates there is a problem with the Thermofluid Pump Frequency Drive in the Boiler control panel. Investigate the problem before restarting.
9. **Boiler Pumps OL:** When illuminated indicates there is an overload on one or both of the Boiler pumps. Investigate the reason for the overload before reset. Overload is in Boiler control panel.
10. **Thermo fluid filling jog:** When selected to "On" the Thermofluid pump is sent a low speed signal from the PLC. This is used when initially filling the system with oil and the heat exchanger is a long distance from the boilers. This fills the pipes to ensure less air and cavitation in the system. Once the system is filled and purged of air, the selection needs to be returned to "Off" so the pump can run at the operating speed set by the PLC.
11. **Thermo fluid Pump Stop:** This selection will Stop the Thermofluid pump.
12. **Boiler Pump Start:** This selection will start the Boiler pumps.
13. **Boiler Pump Stop:** This selection will stop the Boiler pumps.
14. **Nitrogen Off/On:** This selection enables the Nitrogen system when "On" is selected.
15. **Nitrogen On:** When illuminated indicates the Nitrogen system is currently enabled.
16. **Nitrogen Drain:** When illuminated indicates the Nitrogen system is currently not pressurized.

NOTE* The Nitrogen system MUST be enabled and pressurized anytime the Thermofluid system is going to be operating. DO NOT attempt to run or heat the Thermofluid oil without the Nitrogen system enabled and Pressurized to a minimum of 20 PSI.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER'S THERMOFLUID SYSTEM - FILLING INSTRUCTIONS

- * 0CF6018-03 Tortilla Chip Fryer
- * 0HE3000-02 Heat Exchanger
- * 0SM3000-02 Support Module
- * 0PLCF60-001 PIPPING KIT



Use reference "0CF6018-03—P&ID 071824—201625—SO116954—FIESTA—ARTE SANO—3000LBS CHIP FRYER" for valve's location.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

NOTE:

Proceed only if circuit is already PRESSURE tested and ALL the safety devices check and in good working condition.



ONLY a drum air operated piston pump must be use to fill the thermal fluid circuit, with a MAX. flow rate not exceeding 3.7 Gal/Min.



Lawrence Part. # AP011-07

PUMP SPECIFICATIONS

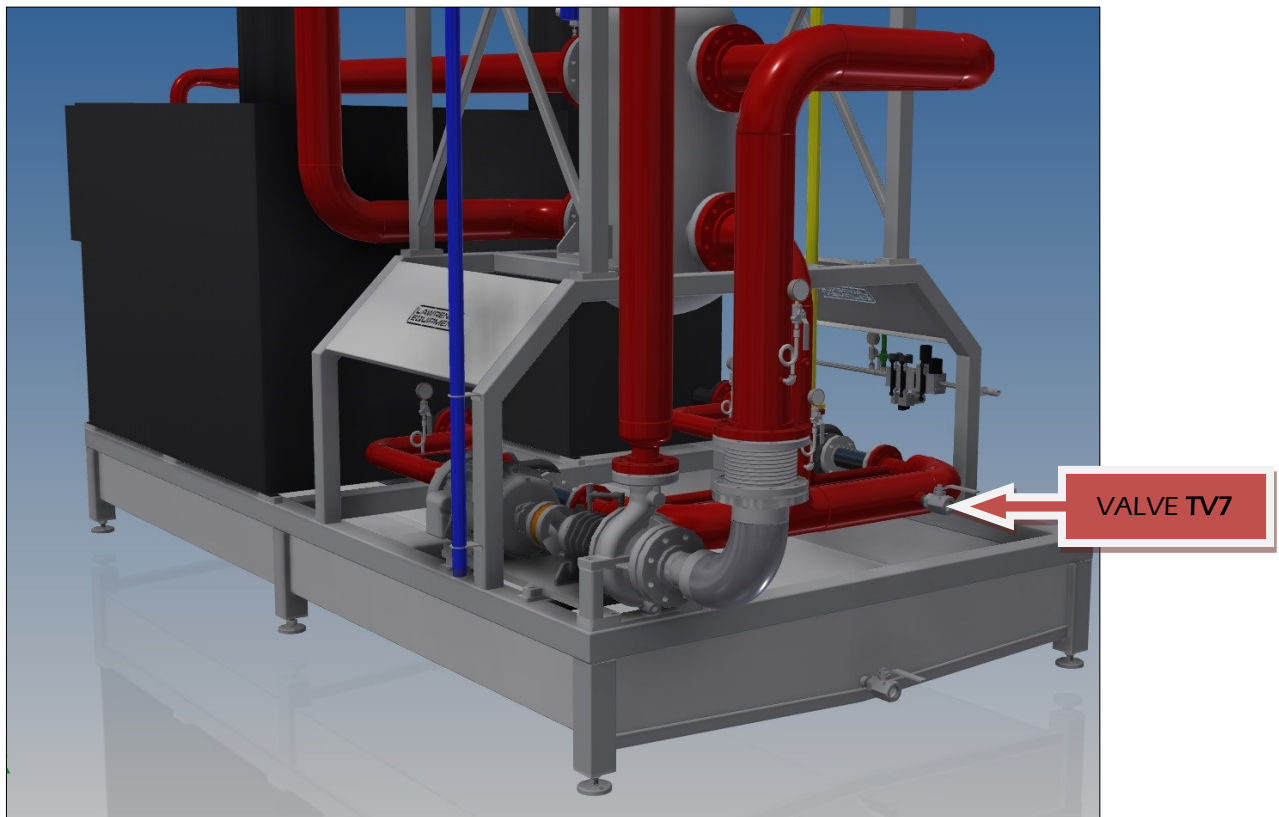
WORKING PRESSURE	30 - 150 PSI
MAXIMUM AIR PRESSURE	150 PSI
AIR INLET CONNECTION	1/4" NPT (F)
PUMP OUTLET CONNECTION	3/4" NPT (F)
NOISE LEVEL	81 db

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

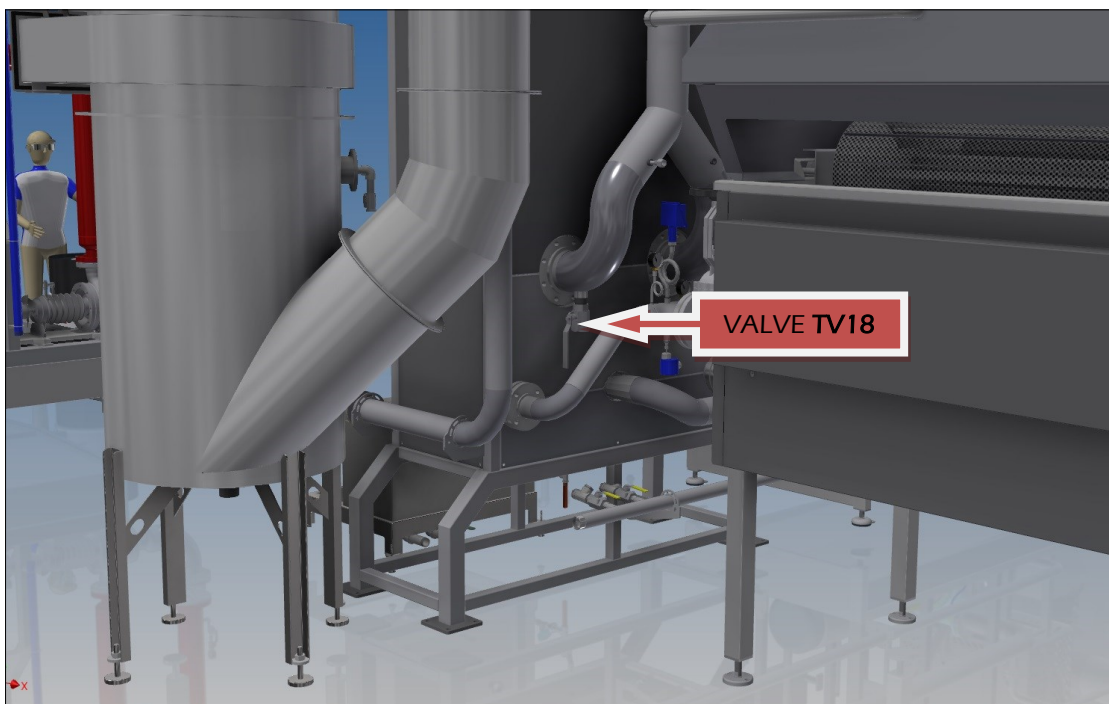
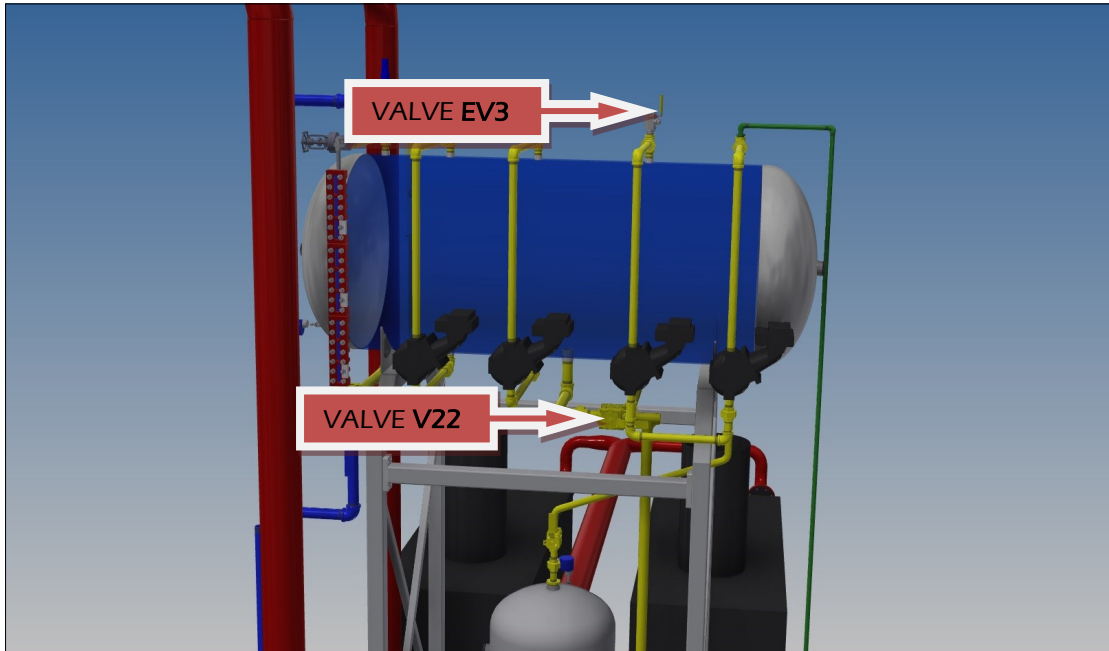
FOLLOW THESE INSTRUCTIONS CAREFULLY TO FILL THE THERMOFLUID SYSTEM

NOTE: USE ONLY MULTITHERM OG-1 THERMOFLUID OIL

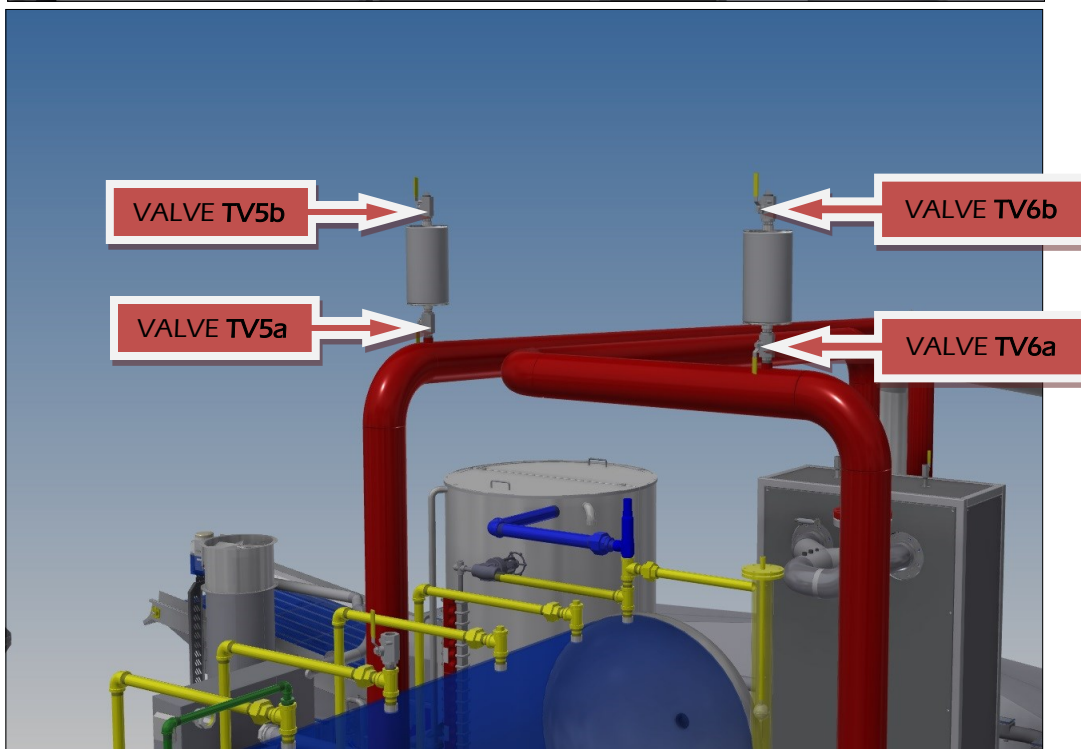
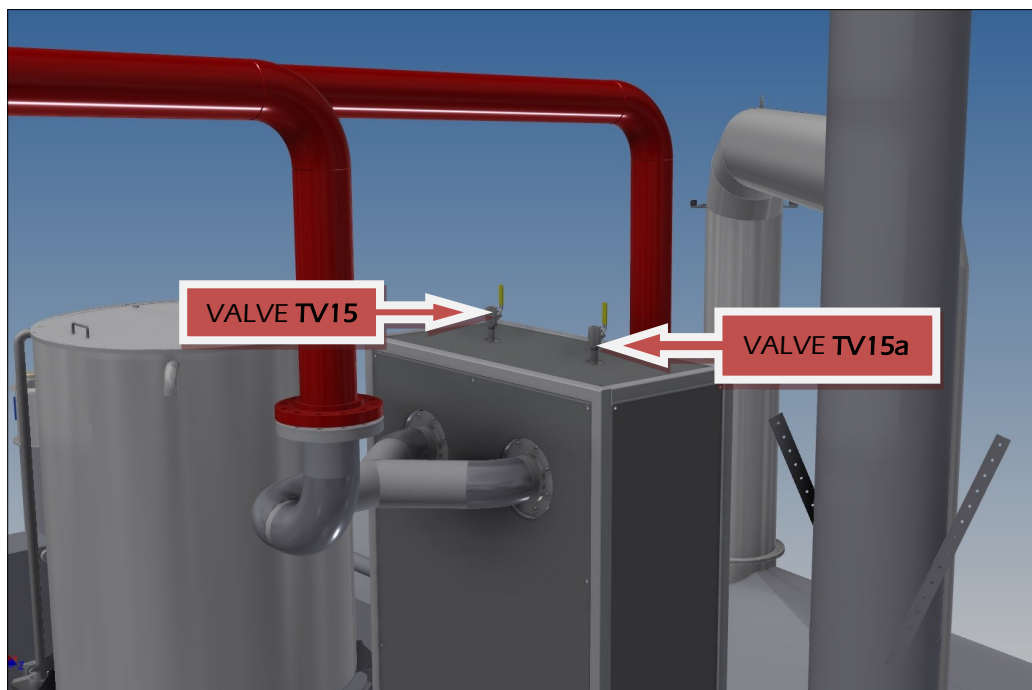
1. Open valves V18, TV5a, TV5b, TV6a, TV6b, V22, EV3, V17, TV7, TV15, TV15a, V28:



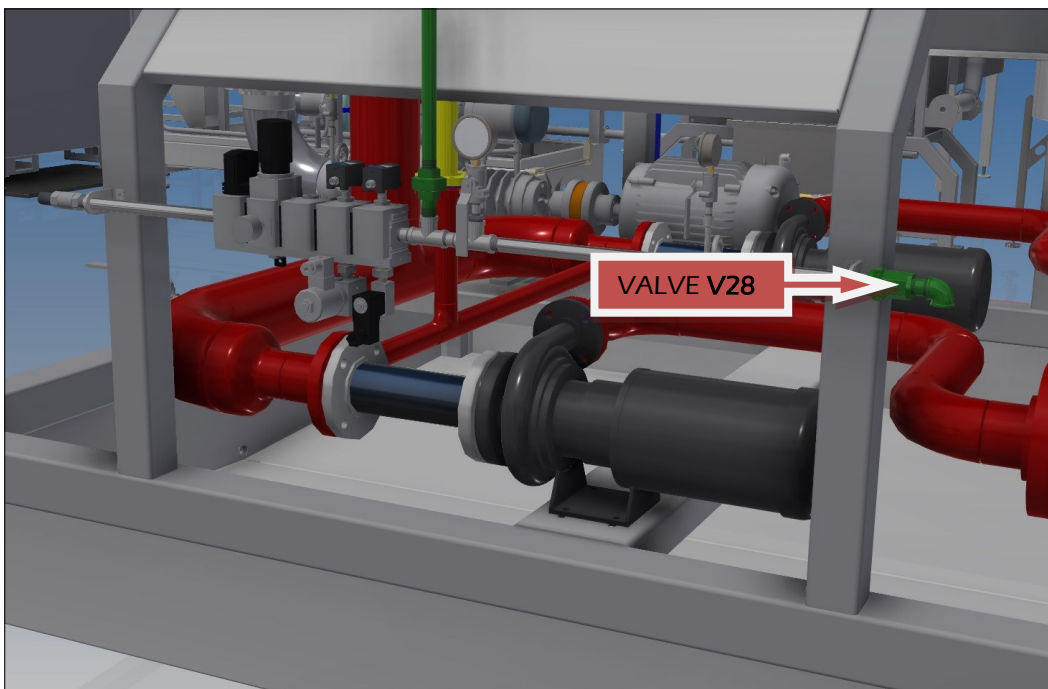
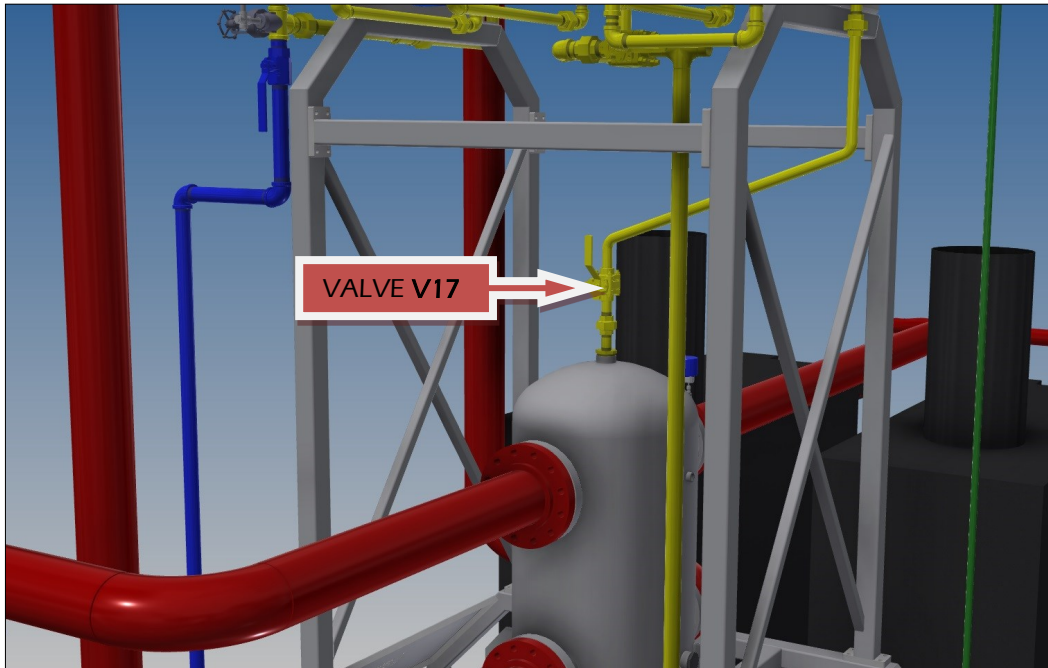
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

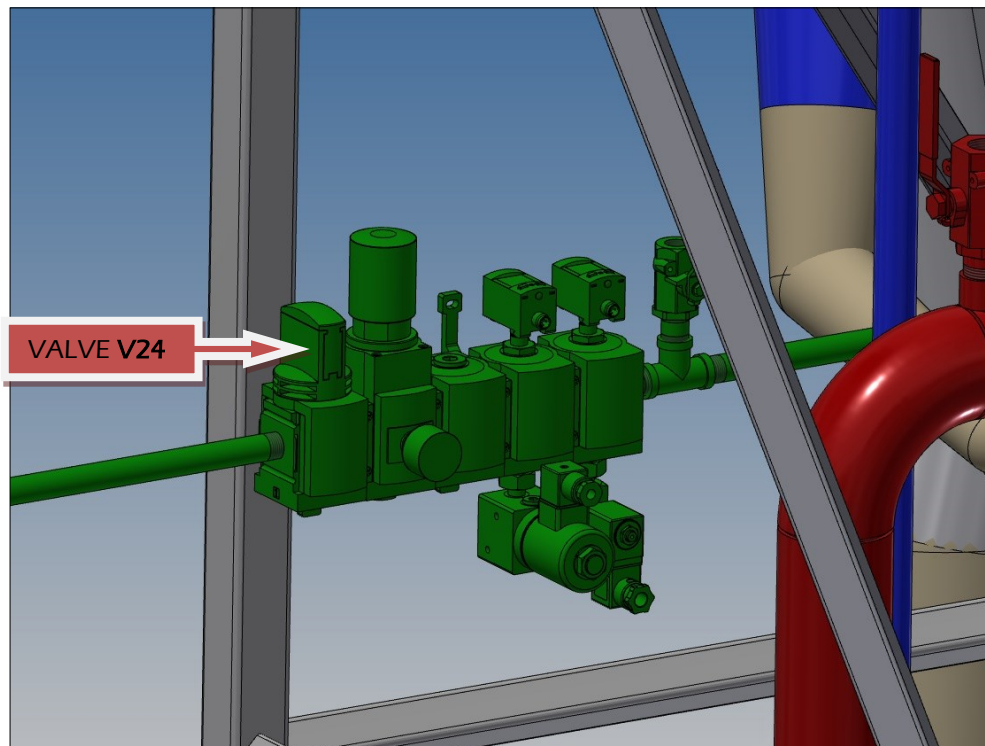
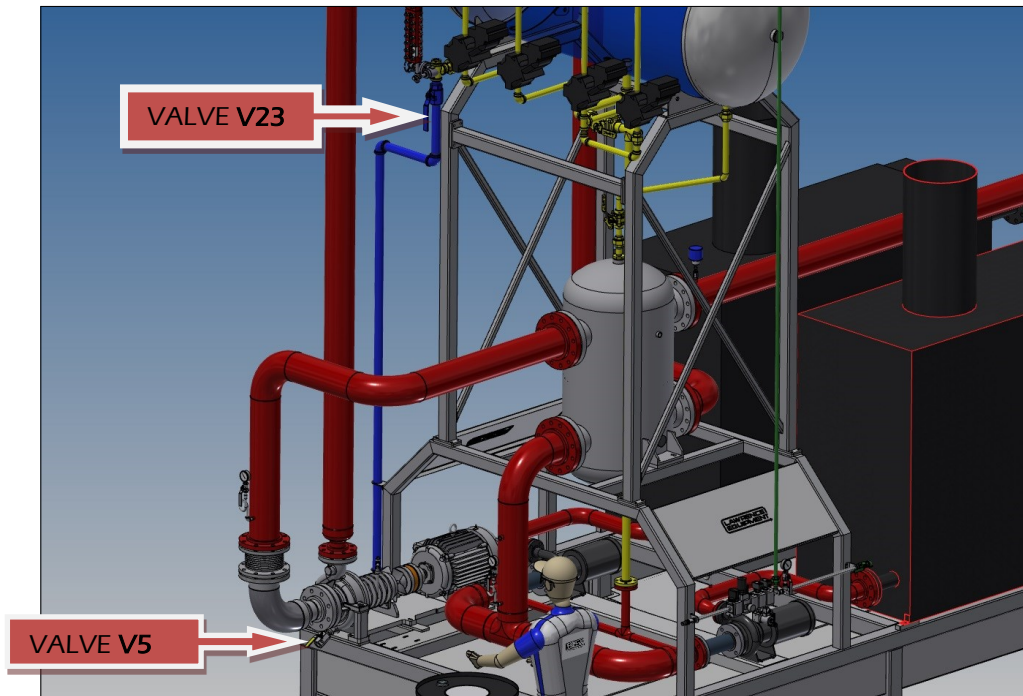


BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

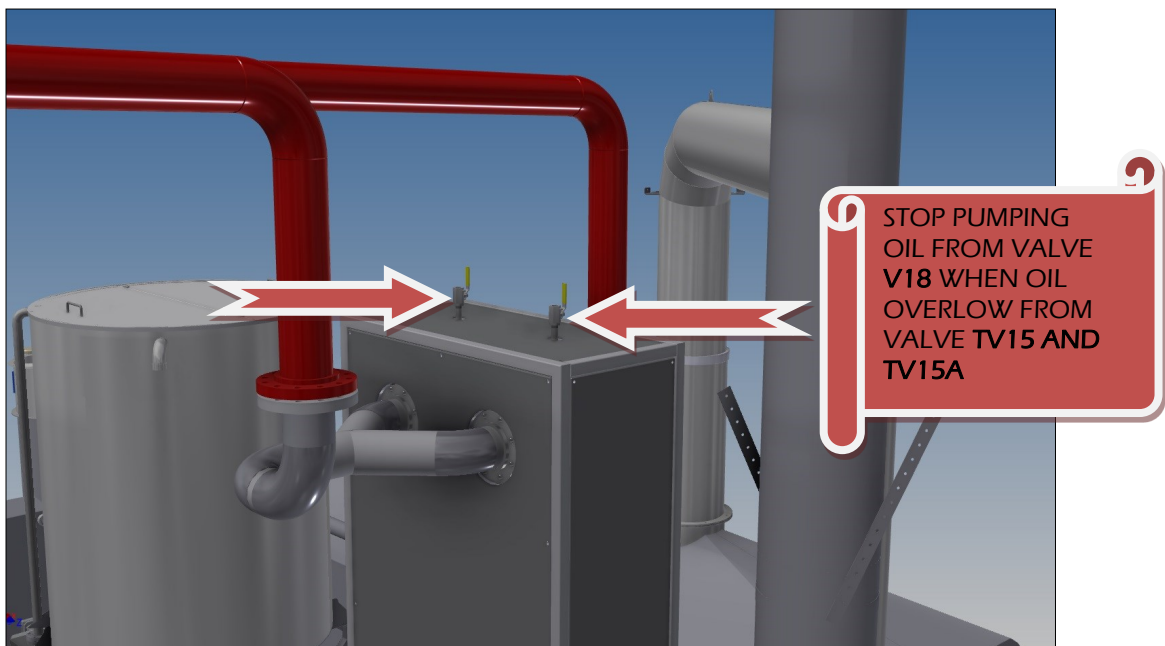
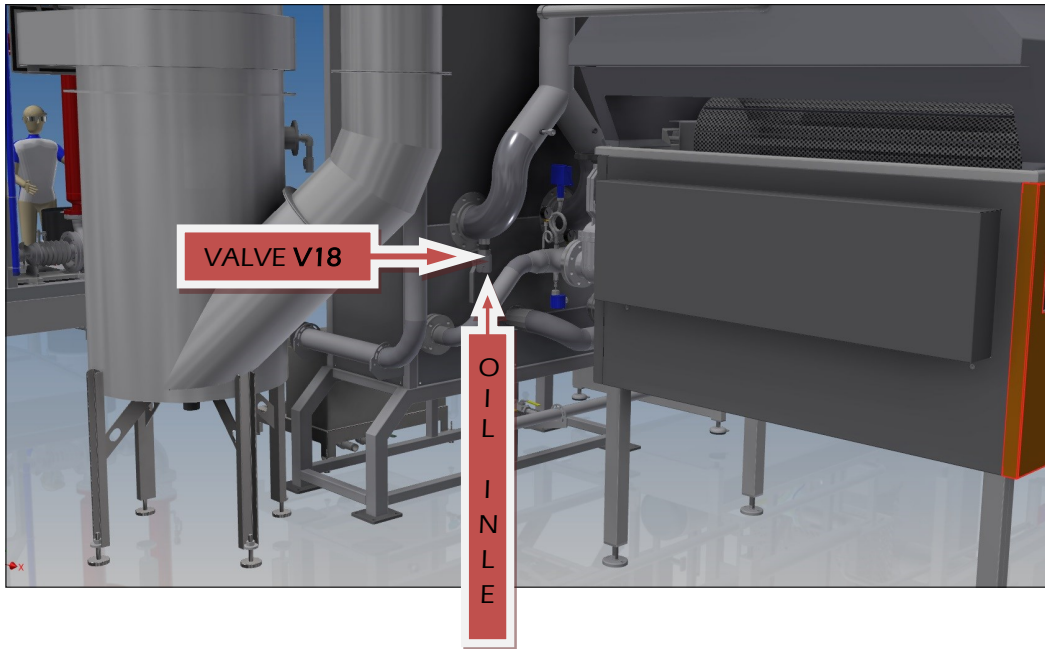
2. Be sure that the following valves are close: V23, V24, V5



Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

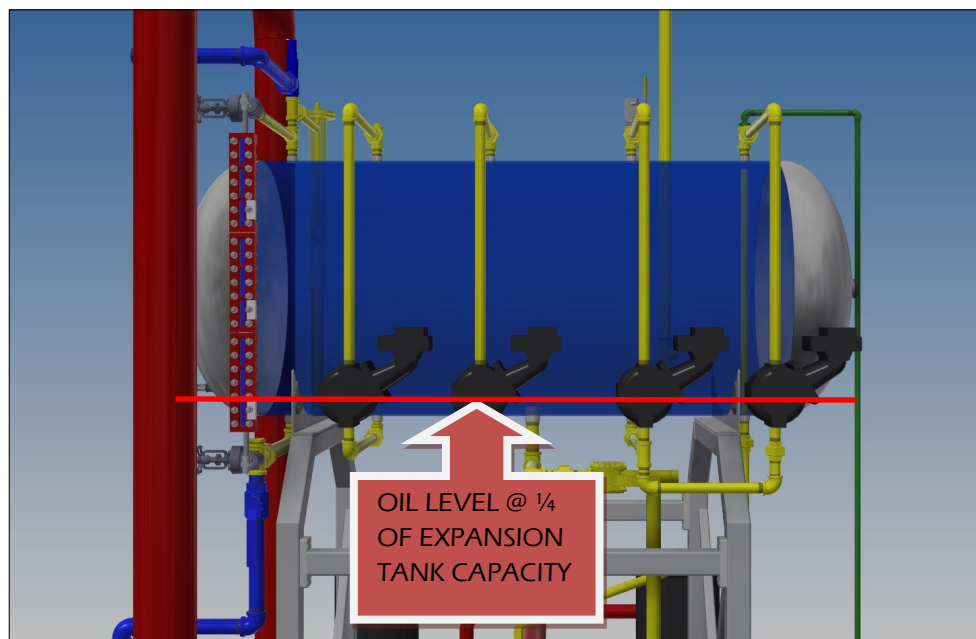
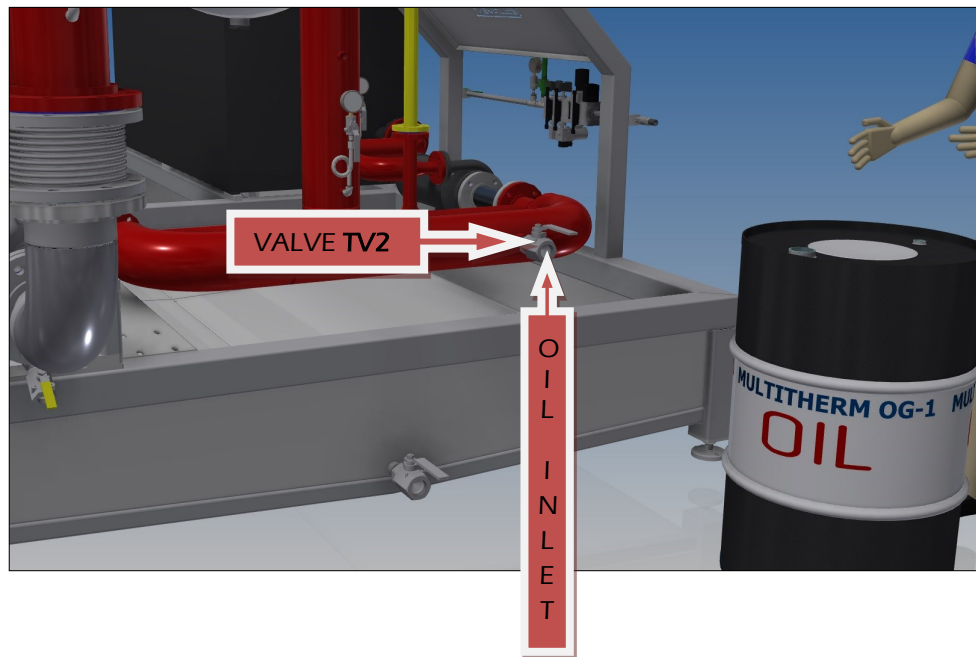
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

3. Connect the filling air piston pump to valve **V18** and start pumping oil VERY slowly in the circuit until oil overflow from valve **TV6**:



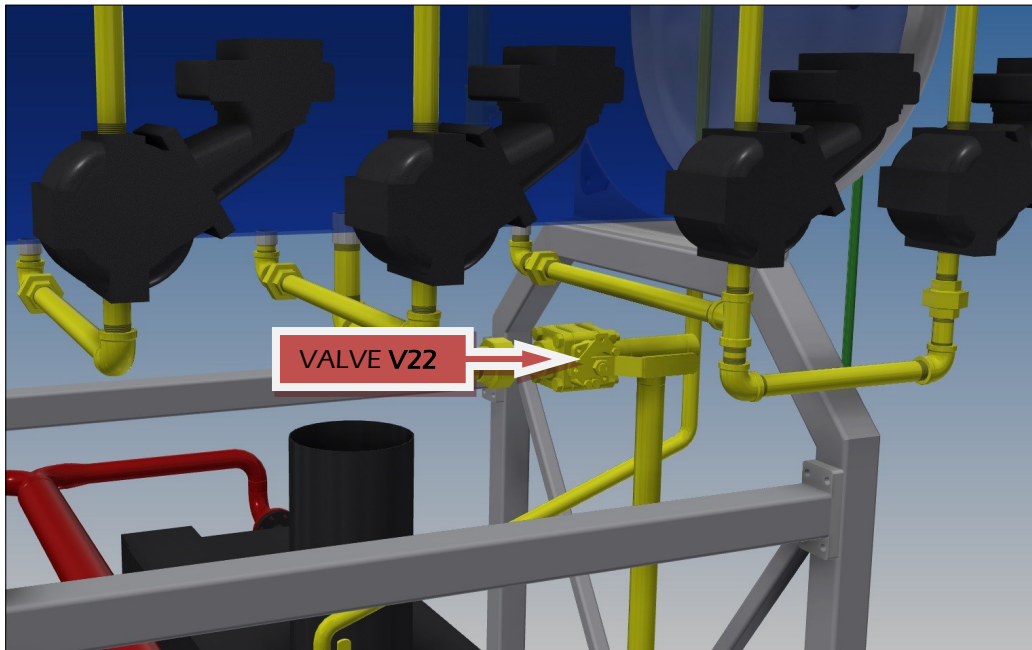
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

4. Close valve **V18**, valve **TV15** and **TV15a** and connect the air piston pump to valve **TV2**. Then start pump oil slowly until level reach $\frac{1}{4}$ of the expansion tank capacity. The level is indicated on the sight glass of the expansion tank by a white band, also, be sure oil level is just above the thermocouple probe and the line indicator on the level switch (NOTE, do not overfill the expansion tank):

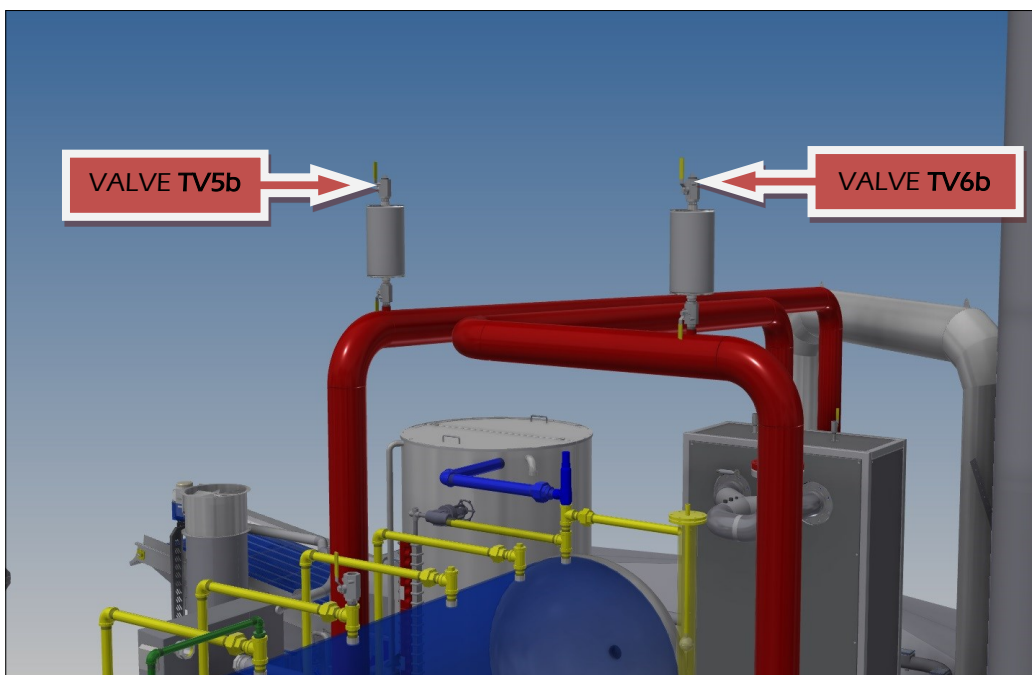


BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

5. Close valve **V22**:



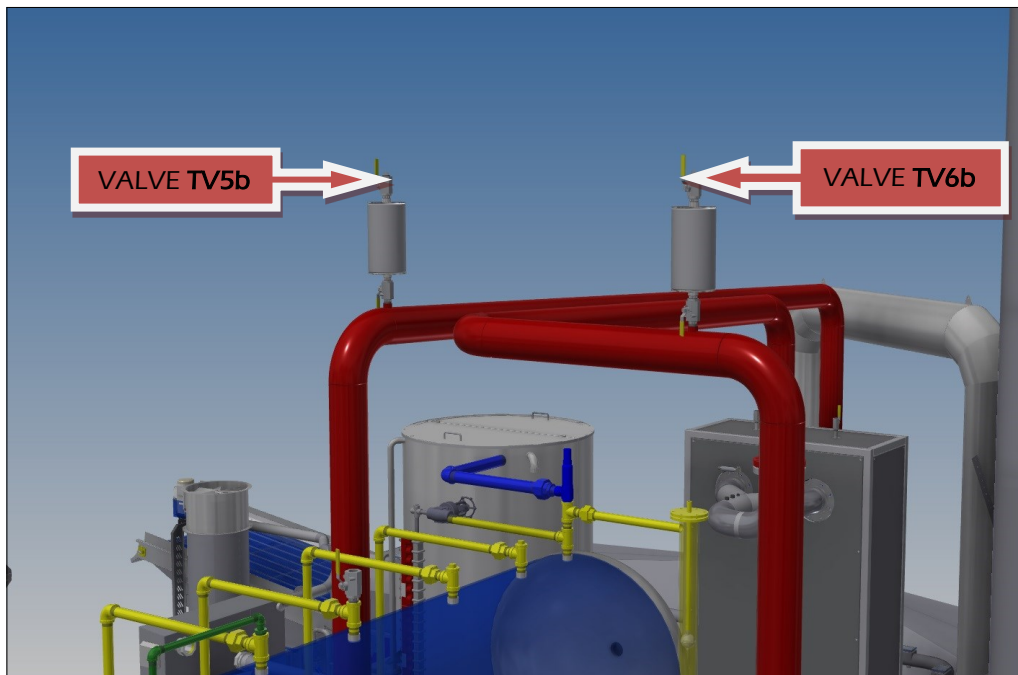
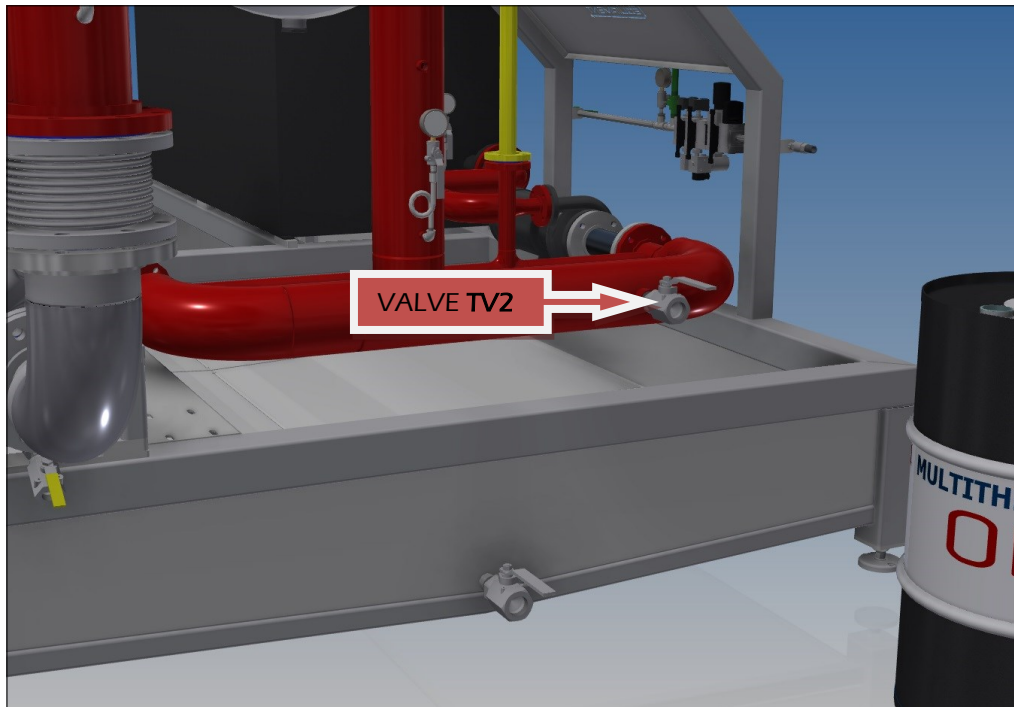
6. After valve **V22** is close, continue to pump oil slowly until oil overflow from the two top **TV5b** and **TV6b** valves. Keep pumping oil until no air bubbling is seen coming out from **TV5b** and **TV6b**.



Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

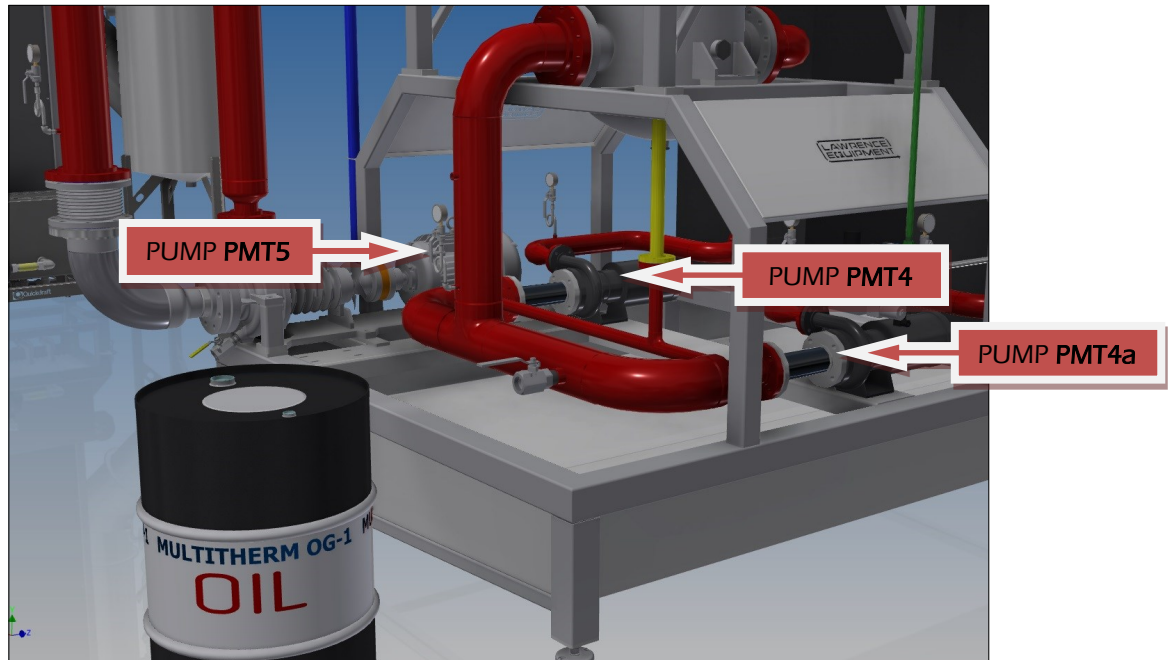
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

7. Stop pumping oil and close valve TV2, TV5b and TV6b:

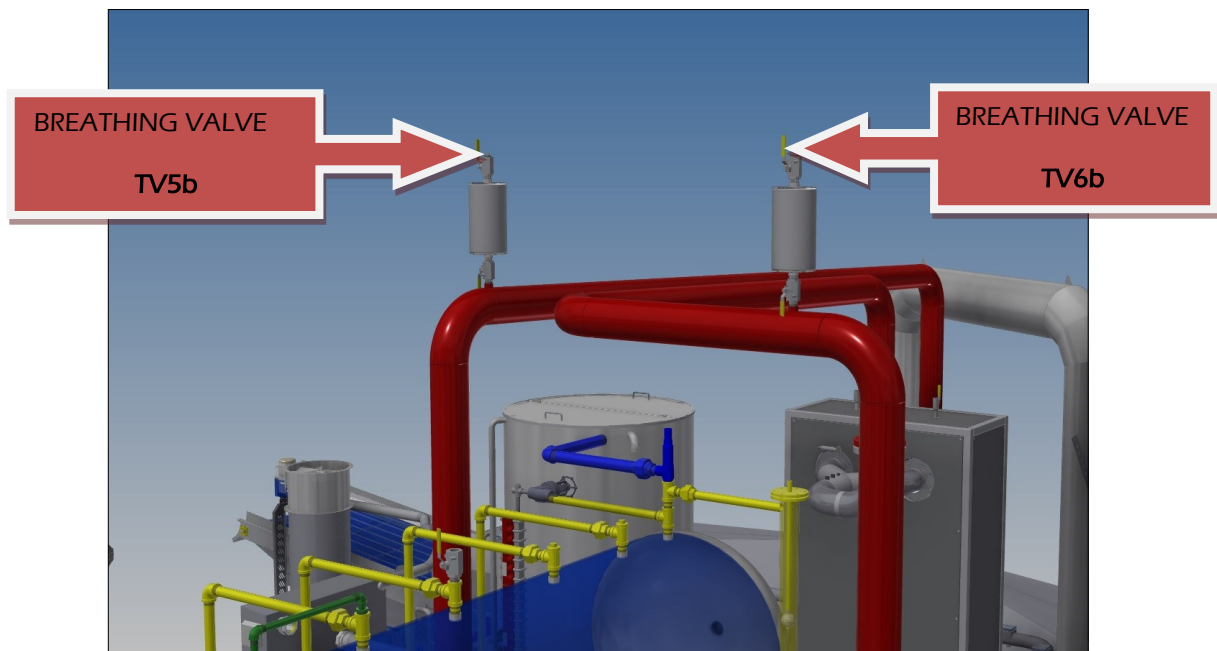


BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

8. Start PUMP **PMT5**, **PMT4** and **PTM4a** and let the system running for 30mins:



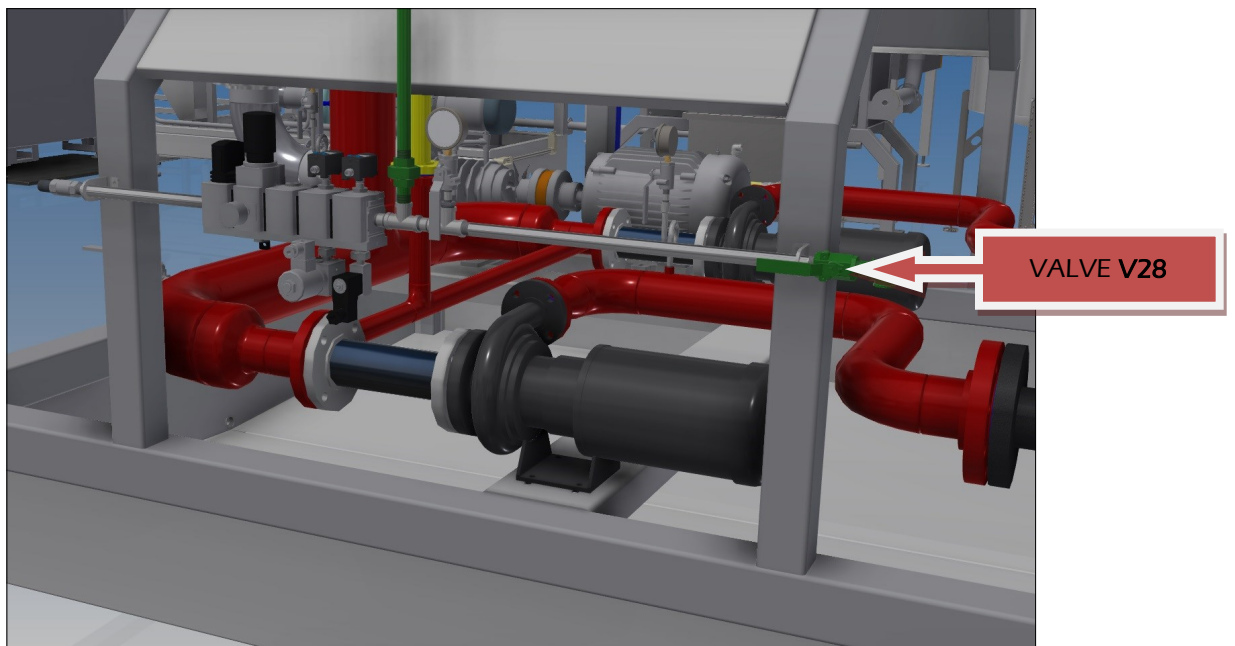
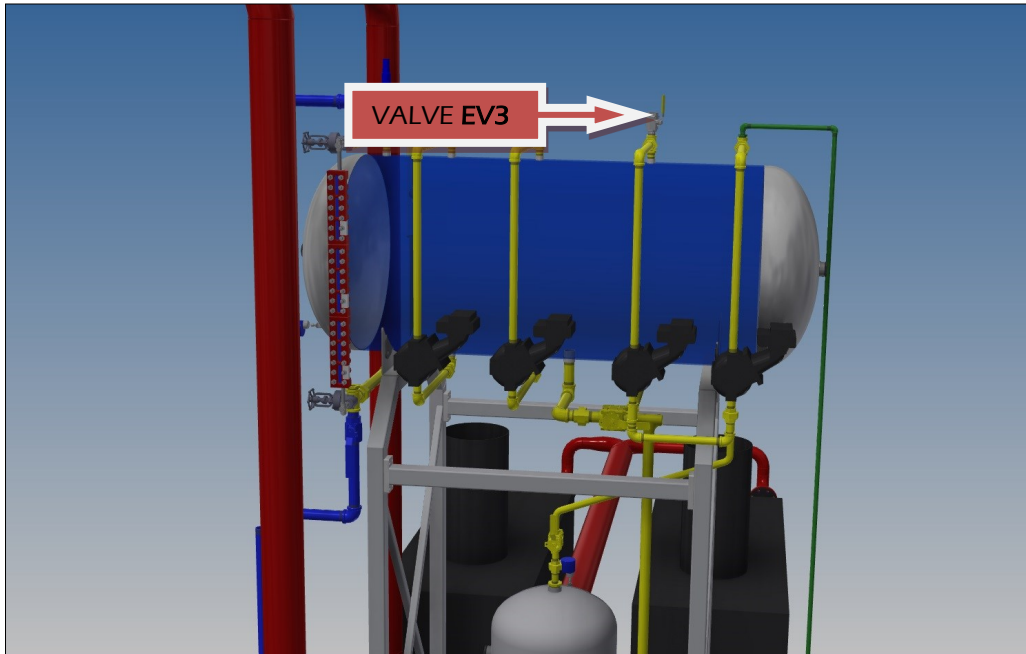
9. Stop all the PUMPS and open the two breathing valves **TV5b** and **TV6b** to purge the circuit from the air collected in the collecting tanks. Keep them open for 5mins and then close them again. Repeat operations 8 and 9 for at least 3 more times.



Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

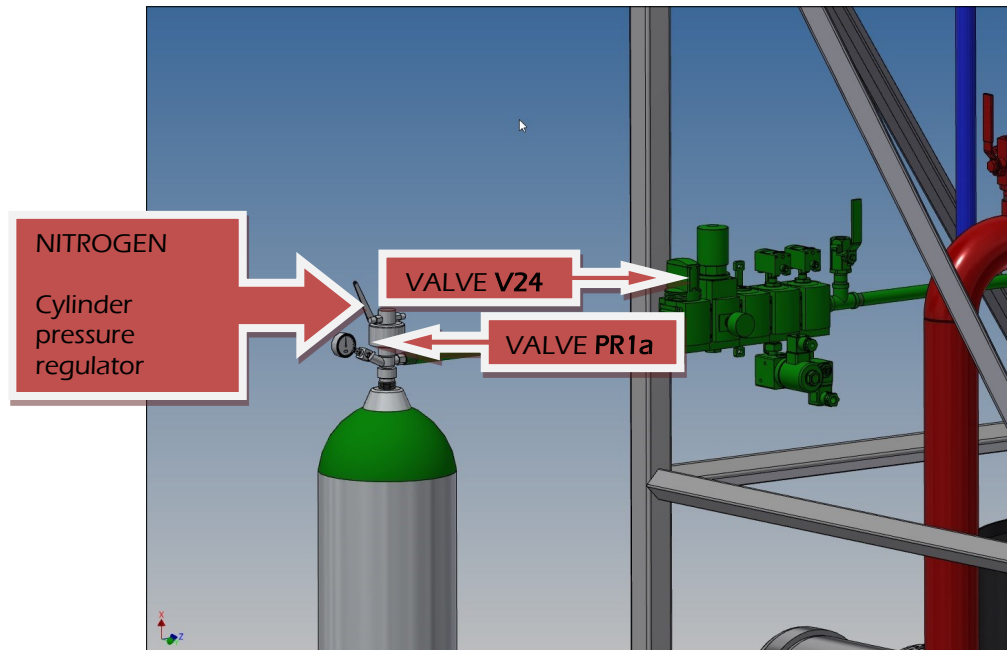
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

10. When all the air breathe out from the circuit, close valve **EV3** and valve **V28**.

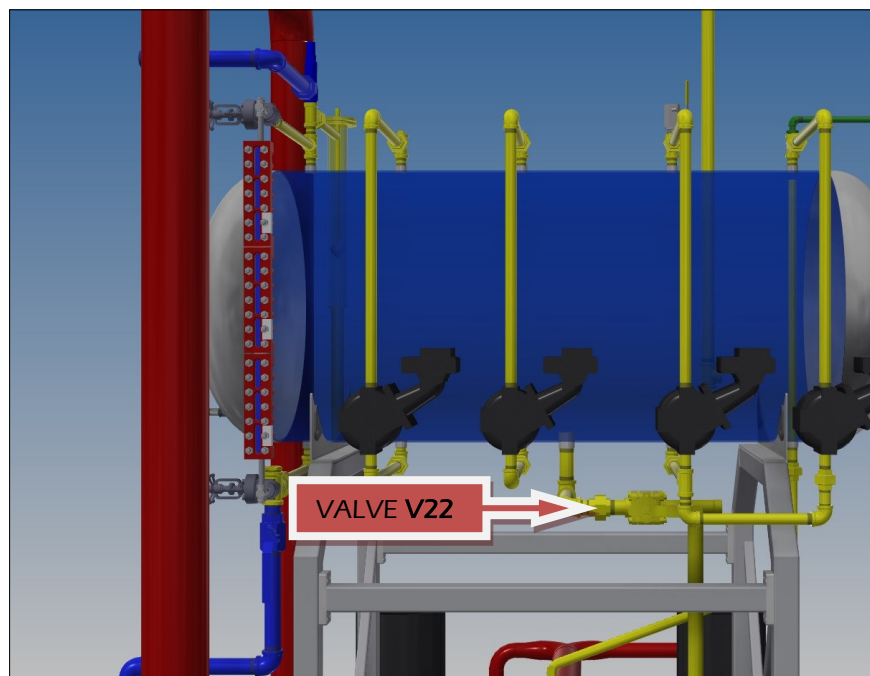


BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

11. Adjust the NITROGEN pressure regulator on the NITROGEN cylinder to **40PSI** meanwhile opening the tank valve **PR1a**, and then the regulator on the FESTO unit to **20psi** meanwhile opening valve **V24**.



12. Open slowly valve **V22** to pressurize the circuit:



Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

13. Check if oil level is still at $\frac{1}{4}$ of the expansion tank (low level switches LSL4 and LSL5 must be close). If level went down that some air is still present in the circuit. Go back to step #4.

If the level remained at $\frac{1}{4}$ then lock all the lockable valves with a lock (ref. valve chart):



The system is now pressurized and ready for the next step.

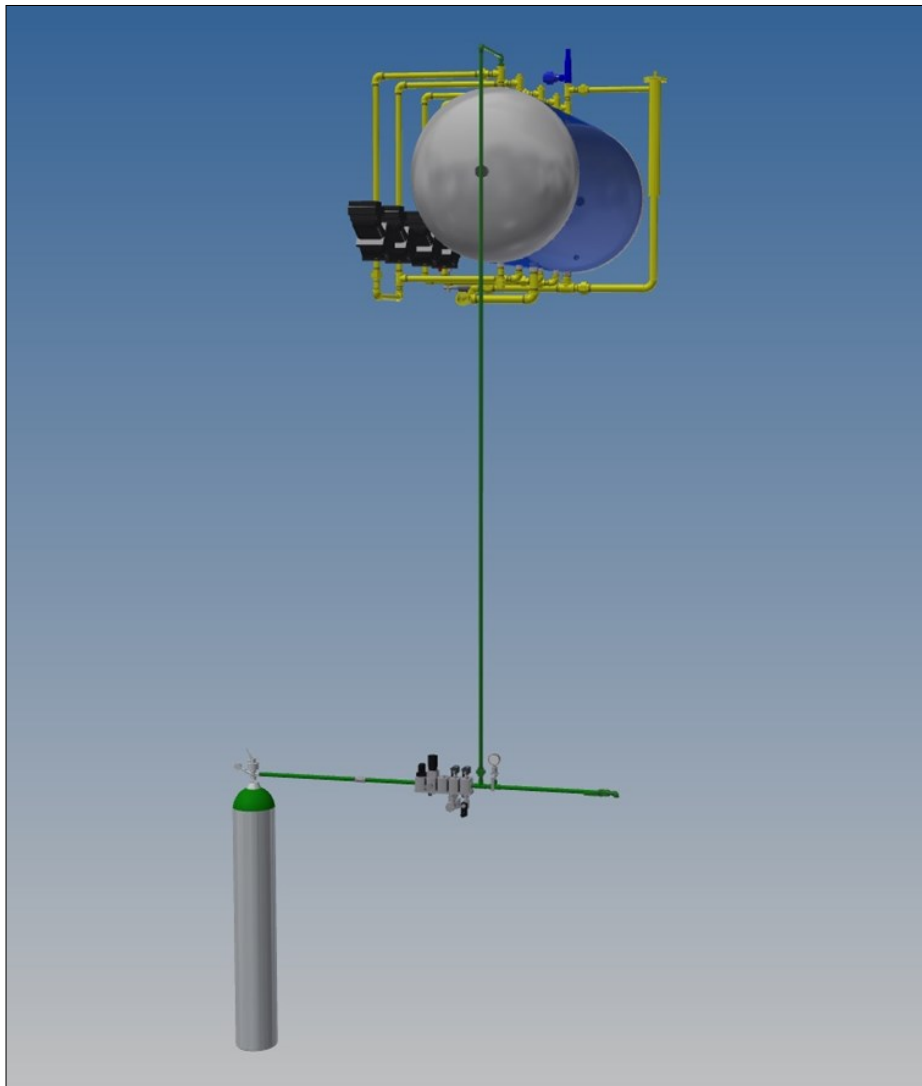
"Heating up the circuit the first time".

Please follow instructions on how to heat up the system the first time and during Startup.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER THERMOFLUID SYSTEM NITROGEN SYSTEM SETUP

0PLCF60-001 PIPING KIT



- * Use reference "0CF6018-03 - P&ID 071824 - 201625 - SO116954 - FIESTA - ARTE SANO - 3000LBS CHIP FRYER" P&ID for valve's location.

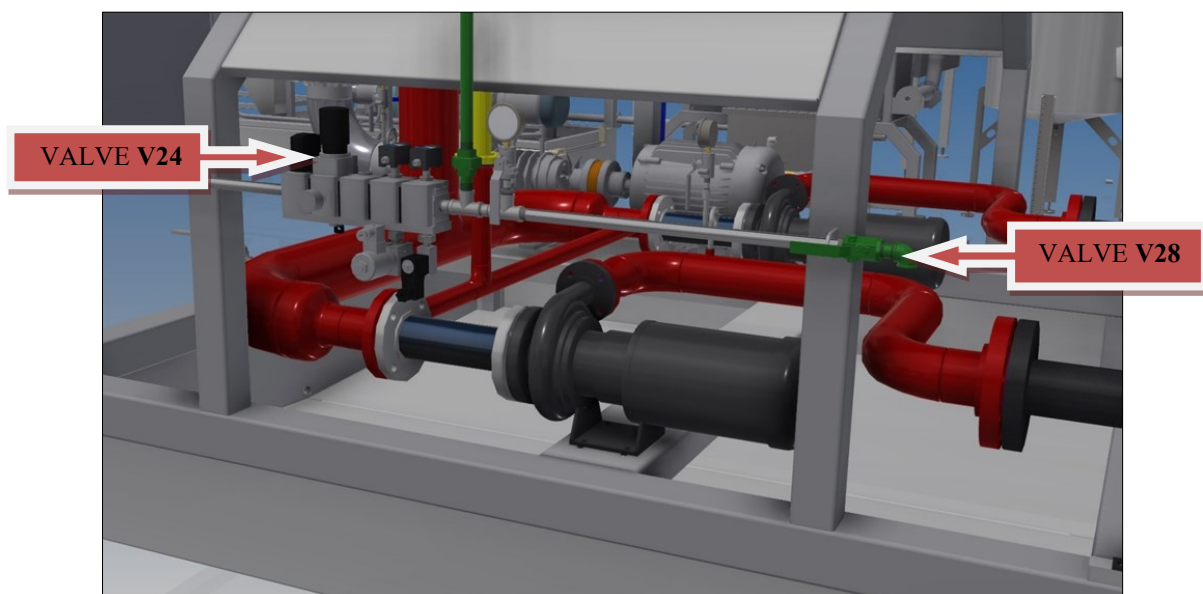
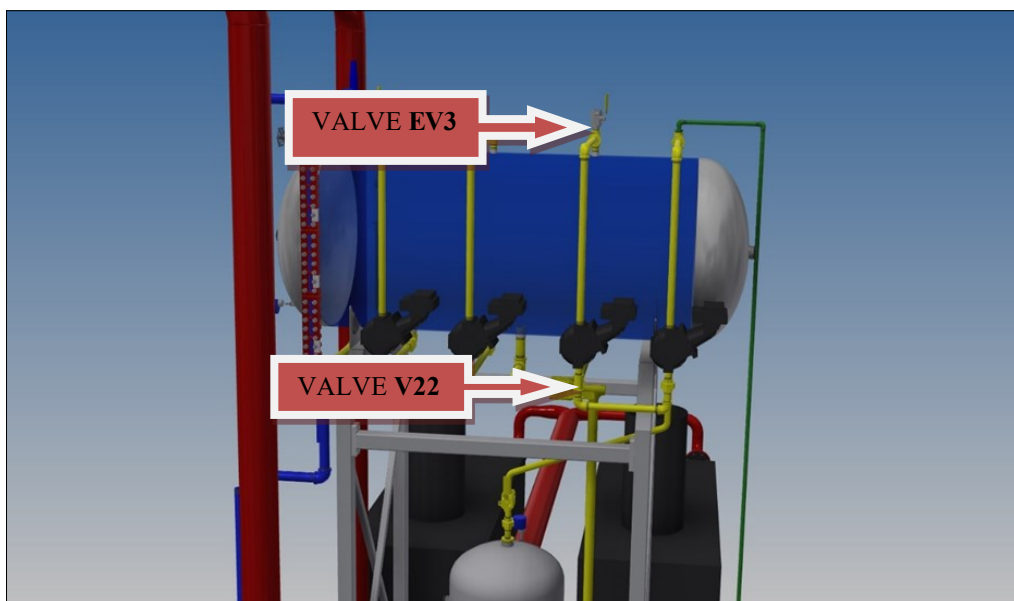
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER THERMOFLUID SYSTEM NITROGEN SYSTEM SETUP

FOLLOW THESE INSTRUCTIONS CAREFULLY TO SET-UP THE NITROGEN SYSTEM.

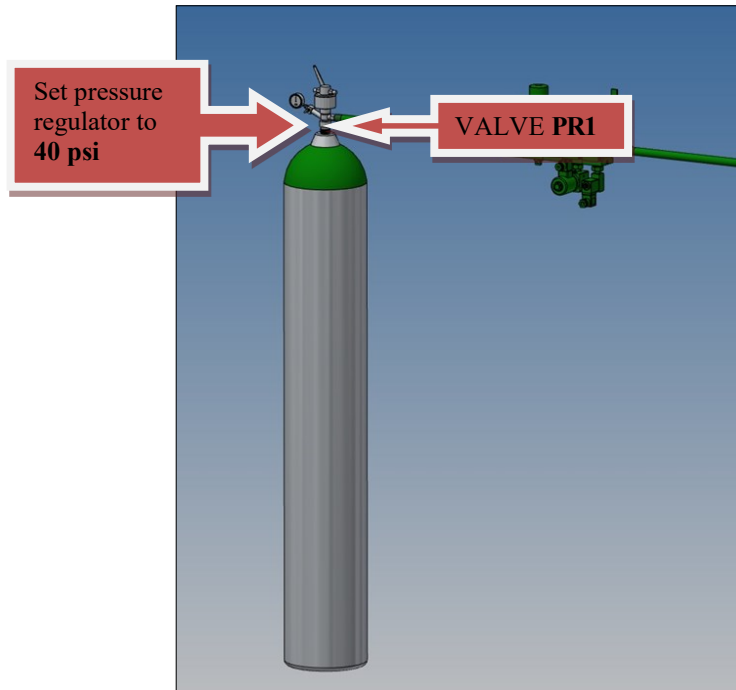
NOTE: SYSTEM MUST NOT BE RUNNING AND OIL MUST BE AT ROOM TEMPERATURE

1. Be sure valve V22, V24 are CLOSED, valve V28 and EV3 are OPEN:

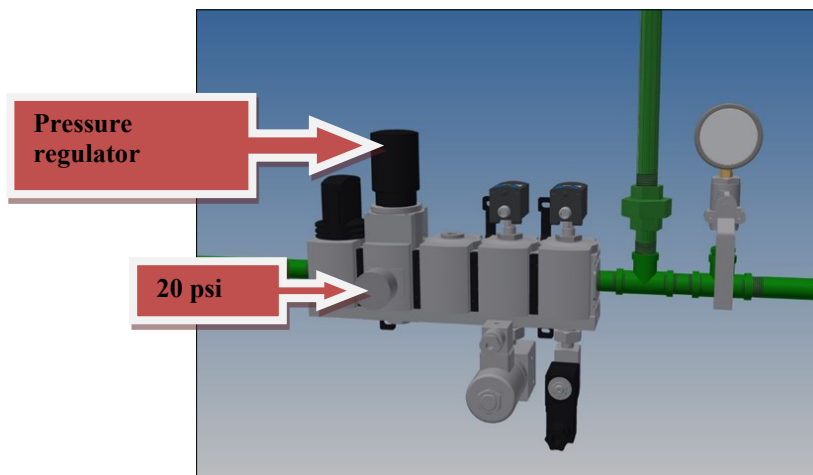


BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

2. Open gently valve **PR1** on the nitrogen cylinder and set the pressure regulator to **40 psi**:

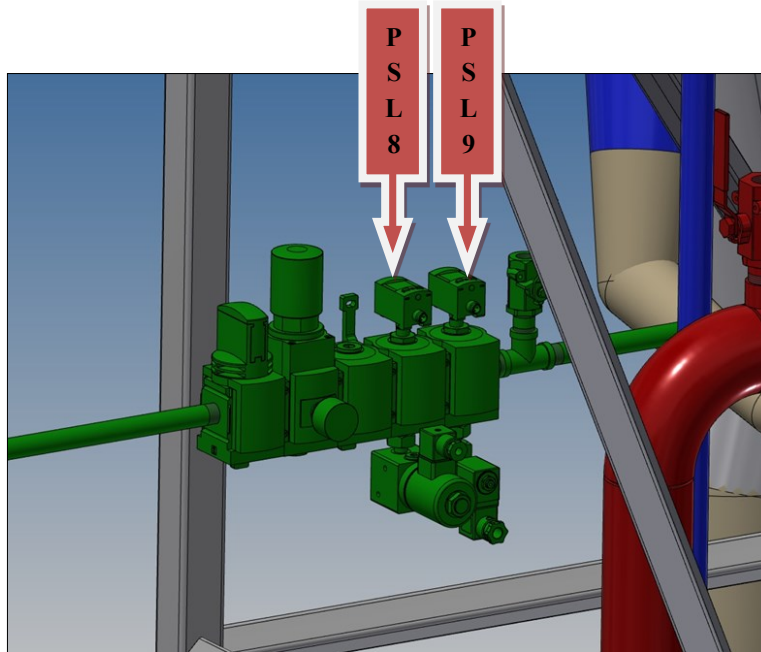


3. Open valve **V24**. Set the pressure regulator on the NITROGEN manifold (part. # AP008-30) to **20psi**:

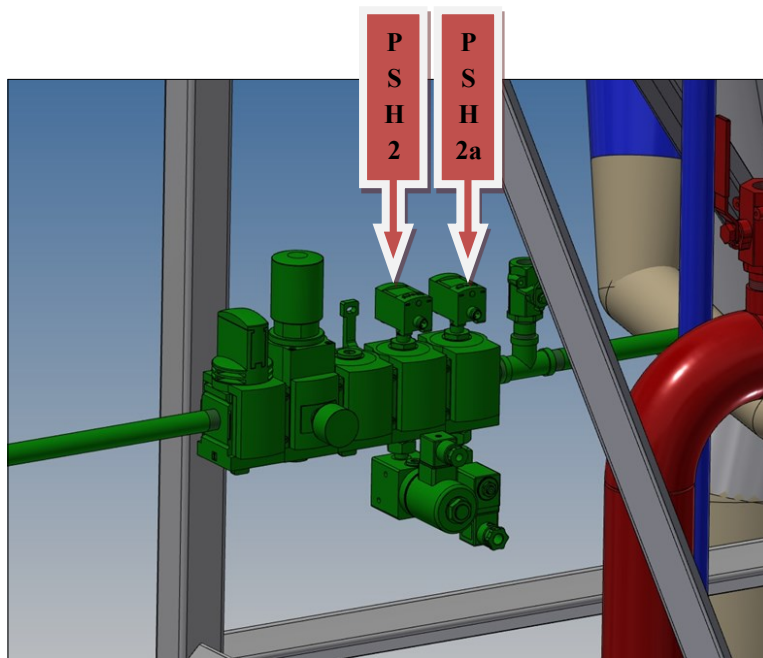


BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

4. Set alarm 1 on PSL8 to 12psi (refer to FESTO manual) and alarm 2 on PSL9 to 20psi.

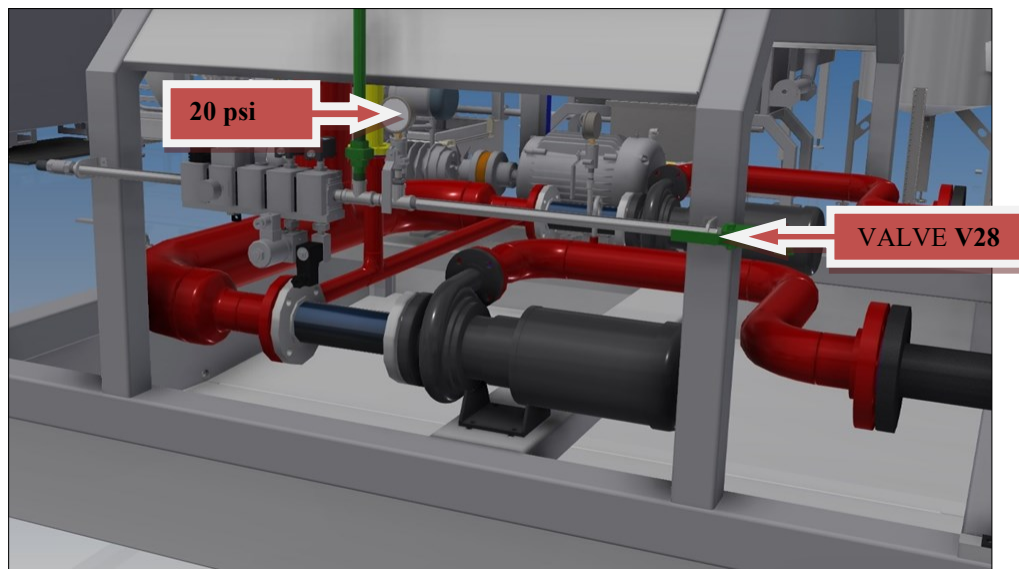
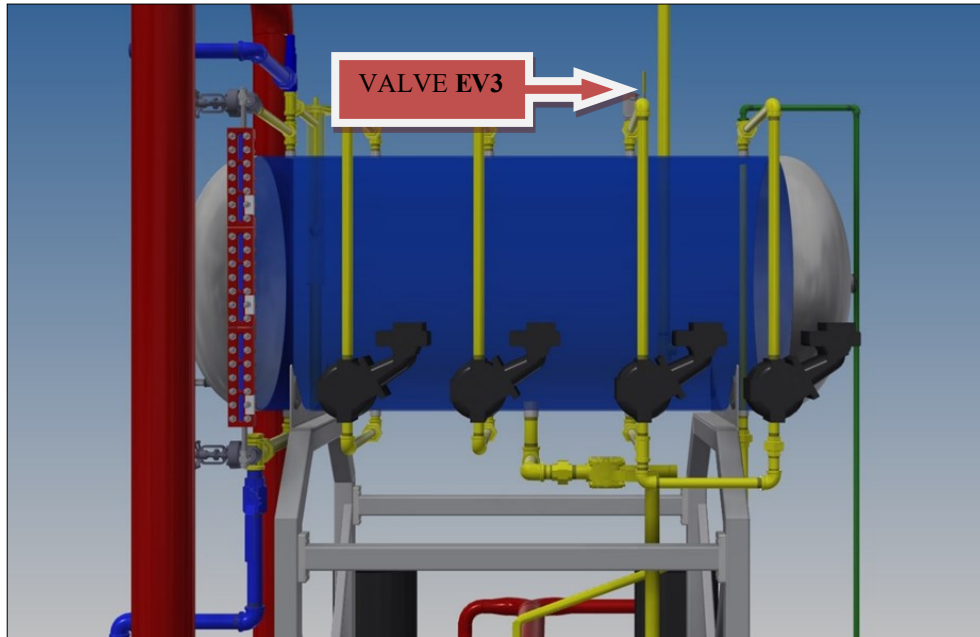


5. Set alarm 1 on PSH2 to 50psi (refer to FESTO manual) and alarm 2 on PSH2a to 55 psi.



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

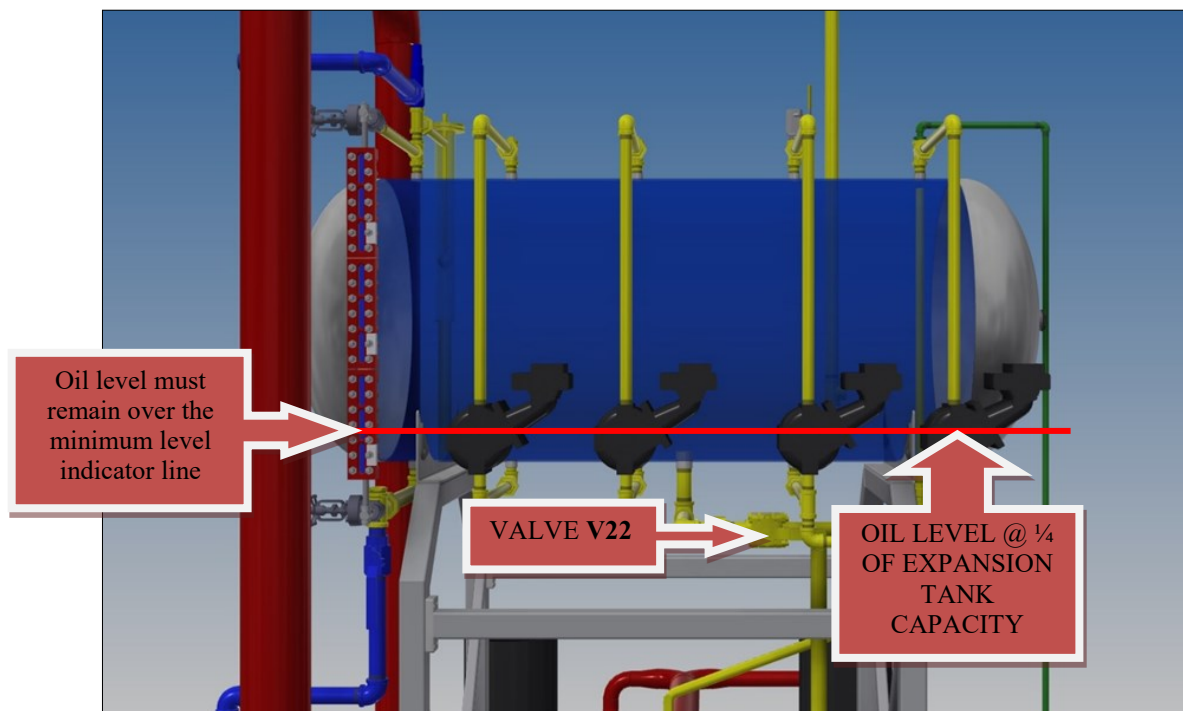
6. Close valve **V28** and valve **EV3** and wait until expansion tank is fully pressurized at **20psi**.



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

7. Open gently valve **V22**. System is now pressurized. Check if the oil level of the expansion tank dropped. If dropped, system still has air in the circuit and **IT MUST BE PURGED AGAIN** until oil level remains stable under pressure.

DO NOT CONTINUE IF OIL LEVEL DROPPED. Refer to Filling Instruction Manual Paragraph #4



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

ALARMS MEANING AND FUNCTIONS

1. **PSL-alarm 1 LOW PRESSURE (<12psi)**: If activated, system shutdown is not able to start until working pressure is restored above 20 psi.
2. **PSL-alarm 2 LOW PRESSURE (from 12psi to 20psi)**: If activated, low-pressure alarm pop up. Circuit verification is needed but still able to run.
3. **Pressure from 20psi to 50psi: Working pressure OK, any alarms should NOT be activated.**
4. **PSH-alarm 1 HIGH PRESSURE (from 50psi to 55 psi)**: If activated, high-pressure alarm pop up. V26 opens and it will try to lower the pressure below 50psi. Circuit verification needed; machine is still able to run, however, a shut down is **STRONGLY** recommended.
5. **PSH-alarm2 HIGH PRESSURE (above 55psi)**: If activated, system shutdown, valve V26 and V27 open trying to lower pressure below 50 psi.

Exceptions:

- a. **IF PSH-alarm 1 is activated and LSH4 is activated, valve V26 remain close.**
- b. **IF PSH-alarm 2 is activated and LSH4 is activated, valve V26 and V27 remain close.**
- c. **IF LS4 is activated, system shutdown, alarm pop up for high level.**
- d. **IF TE3 is above 180F, system shutdown, alarm pop up for high temperature.**

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

THERMO FLUID FRYER START UP & SHUTDOWN

Before attempting to start the fryer, ensure that all water has been removed from the system. There should be no water in any tanks, kettles, pump or piping of the system. Any water left over from cleaning/CIP functions can and will explode if the oil is introduced and heated above 210 degrees Fahrenheit. A STRICT “boil out” procedure is to be followed which is explained further in this manual. Follow this “boil out” every time the fryer is started up after a cleaning or water rinse.

NOTE ** DO NOT RUN FRYER OIL PUMP DRY! SEVERE DAMAGE WILL OCCUR!!**

STEP 1: DRAINING, FILLING AND STARTING THE FRYER EQUIPMENT.

1. Before adding any cooking oil, ensure the system is drained of all water.
2. Turn On power and Air to the system.
3. Enable the safety system on both electrical enclosures.
4. If it is up, Lower the fryer kettle hood all the way down.
5. Open valve V8 (cooking oil pump, low point drain)
6. Open valve V13 and V14 (cooking oil heat exchanger drains)
7. Check the oil storage tank, ensure no water is inside.
8. Turn On the fryer “Auto Fill” and allow it to open the valve under the oil storage tank. Ensure all water drained.
9. Open valve V3 (CIP hood spray nozzles)
10. Check the fryer kettle, ensure no water in it.
11. Check the fryer drum filter, ensure no water.
12. If no water is draining, close all the valves previously opened.
13. Follow the Valve Chart, set the valves to the orientation used for “cooking oil filling”.
14. Check all auxiliary machinery, ensure all conveyors are ready, covers installed, hoses connected.
15. Turn On the fresh oil supply pump.
16. Ensure the fryer “Auto Fill” function is turned On.
17. Allow the system to fill with cooking oil. The oil storage tank should fill the fryer kettle plus fill the oil storage tank up to the low level float.
18. From the fryer touch screen go to the “Start Screen” then the “Fryer Screen”.
19. Select “System Start” to start the system. When using “System Start”, this turns on the fryer pump, the filter box, the fryer paddle and drum conveyors, the discharge, the aspiration conveyor, the aspiration blower, the exhaust blower and the fryer oil auto fill function.
20. Double check all valves/drains to ensure no drains leaking and proper orientation of the valves.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

THERMO FLUID FRYER START UP & SHUTDOWN (continued)

Before attempting to start the fryer, ensure that all water has been removed from the system. There should be no water in any tanks, kettles, pump or piping of the system. Any water left over from cleaning/CIP functions can and will explode if the oil is introduced and heated above 210 degrees Fahrenheit. A STRICT "boil out" procedure is to be followed which is explained further in this manual. Follow this "boil out" every time the fryer is started up after a cleaning or water rinse.

STEP 2: STARTING BOILERS AND WATER BOIL OUT

1. The fryer system should be running, the oil pump running, the oil circulating at level in the kettle at this point.
2. Turn on both of the Thermofluid boilers Main gas supply valves.
3. Start the Boilers by switching the control power toggle switch to ON and the burner toggle switch to ON.
4. Use the Down arrow on the boilers temperature controller to set the temperatures to 225 degrees Fahrenheit.
5. From the Thermofluid control panel, start the Boiler pumps and thermofluid pump.
6. Set the cook oil temperature on the Fryer control touch screen to 215 degrees Fahrenheit.
7. The controller on the boilers will take a few minutes to purge, light the pilot and ignite the Main burner. If any faults occur during this time, the controller will fault and display an alarm. When this initially happens, Simply reset the controller on the burner control box. If the burner does not light after three tries, notify supervision or maintenance.
8. Once the Boilers are lit and heating up, look for any leaks or signs of problems. Its easier to fix a leak or issue before the cook oil and Thermofluid are at full running temperature.
9. When the oil temperature gets close to 212 degrees Fahrenheit, the water residual (if any) will begin to boil and turn to steam. This is the "Boil Out" where the oil is heated enough to turn the water to steam.
10. Allow the system to continue to run at 215 degrees Fahrenheit for 45 minutes. **DO NOT** raise the oil temperature any higher than this at this point!!!
11. After 45 minutes of running at 215 degrees Fahrenheit, raise the cook oil temperature to 250 degrees Fahrenheit and the Thermofluid boilers temperature to 280 degrees Fahrenheit. Continue to monitor for signs of steam flashing. If there are pops from steam in the heat exchanger, lower the COOK oil temperature to 215 degrees Fahrenheit and run another 30 minutes of boil out.
12. Once the boil out is complete, Raise the Cook oil temperature to the running temperature and set the Thermofluid boilers to 515 degrees Fahrenheit.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

THERMO FLUID FRYER START UP & SHUTDOWN (continued)

Before attempting to start the fryer, ensure that all water has been removed from the system. There should be no water in any tanks, kettles, pump or piping of the system. Any water left over from cleaning/CIP functions can and will explode if the oil is introduced and heated above 210 degrees Fahrenheit. A STRICT “boil out” procedure is to be followed which is explained further in this manual. Follow this “boil out” every time the fryer is started up after a cleaning or water rinse.

STEP 3: END OF PRODUCTION

1. When the production run has finished, start by turning Off the Infeed belt after the last of the product has gone into the fryer kettle. Navigate to “Maint Screen” and select “Infeed Belt” to turn it Off.
2. Turn off the Thermofluid boilers by toggling the “burner” toggle switch to “OFF” on both boilers. .
3. On the Fryer touch control panel, Navigate to the “Maint Screen” and turn Off the “Oil Auto Fill” system.
4. Once all the product has past the Aspiration conveyor, turn Off the Aspiration conveyor and Blower. Navigate to “Maint Screen” then select “Aspiration Conveyor” and “Aspiration Blower” to turn Off.
5. Allow the filter, oil pump, paddles and discharge to continue running. This will allow the oil and Heat Exchanger to cool down evenly. NEVER shut Off the Cooking oil pump when the boilers are running, this may overheat and burn the cooking oil. This will not harm the thermofluid oil.
6. Once the Fryer cooking oil is at or below 200 degrees Fahrenheit, the oil can be pumped into either the storage tank above the filter assembly or into an external storage tank.

NOTE: If the production is done for the day and the system (including boiler) is getting shut down, Allow the Thermofluid boiler to run with NO HEAT for one full hour OR until the Thermofluid is at or below 150 degrees Fahrenheit.

7. Navigate to “Maint Screen”, select “Fryer System OFF” to turn Off the system.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

THERMO FLUID FRYER START UP & SHUTDOWN (continued)

Before attempting to start the fryer, ensure that all water has been removed from the system. There should be no water in any tanks, kettles, pump or piping of the system. Any water left over from cleaning/CIP functions can and will explode if the oil is introduced and heated above 210 degrees Fahrenheit. A STRICT “boil out” procedure is to be followed which is explained further in this manual. Follow this “boil out” every time the fryer is started up after a cleaning or water rinse.

STEP 4: REMOVING COOKING OIL FROM THE SYSTEM

1. To pump into the storage tank, stop the system. Navigate to “Fryer Screen”, select “Fryer system Stop”. This will stop the oil pump, paddles, discharge, etc.
2. Ensure the storage tank is empty or no higher than the low level float switch. This will ensure the oil left in the kettle, piping and filter box will fit in the storage tank.
3. Ensure Valve V7 (cooking oil storage tank) or V4A (Cooking oil used oil tank) is OPEN.
4. Ensure Valve V8 (cooking oil main drain) is CLOSED.
5. Navigate to the “Maint Screen” then select “Drain Pump”. Turning the Drain pump On will pump the oil FROM the kettle and filter system TO the storage tank.
6. Monitor the oil level by looking down into the filter box. The oil is fully drained when no oil can be seen in the elbow at the bottom of the box.
7. To ensure the system is fully drained of oil, OPEN Valve V8 (cooking oil pump drain) to drain off the last gallon or two of oil.
8. Once the system is completely drained of oil, CLOSE V7, V14, V15 and V17.
9. To Pump into an external storage or disposal tank, Stop the system. Navigate to “Fryer screen”, select “Fryer System Stop”. This will stop the oil pump, paddles, discharge, etc.
10. Ensure Valve V7 (cooking oil storage tank) is CLOSED.
11. Connect a hose or pipe from the line connected to V8 (cooking oil main drain) to an external tank or disposal container then OPEN V8.
12. Navigate to the “Maint screen” then select “Drain Pump”. Turning the Drain Pump ON will pump the oil FROM the kettle and filter system TO the external tank or disposal container.
13. Once the system is empty, stop the drain pump, Close V8, remove the hose from Drain V8. Place a small container under the drain opening for V8, open V8 to ensure the oil residue is all drained out.
14. Once the system is completely drained of oil, Close V8,

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER START UP PROCEDURES

*** Ensure the system does not contain any water or CIP chemicals. Failure to drain according to instructions will cause severe personal injury and mechanical damage.**

1. Do a walk around of the equipment, look for covers out of place, doors removed, chutes removed and any motor disconnect switches off (investigate why they are shut off before turning on). Replace or install any guards, doors, covers, chutes, etc. that have been removed. DO NOT operate with safety panels or doors removed.
2. Ensure the compressed air is connected (if removable hose) and the manual valves supplying air to the Thermofluid Frying System and the Fryer Support module are all open.
3. Ensure the electrical disconnect on the fryer panel is turned on.
4. Ensure the electrical disconnect on the boiler panel is turned on.
5. Energize the safety systems on both Boiler panel and the Fryer panel.
6. Ensure the Nitrogen tank is full and connected, all manual valves are open, nitrogen drain valves closed and there is at least 20 PSI of Nitrogen pressure in the expansion tank. *(See Fig. A)*

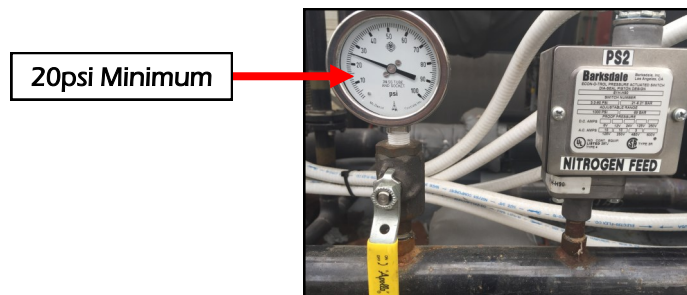


Fig. A

7. On the Boiler control panel, select "Start" button to turn on boiler pumps. *(See Fig. B)*
8. On the Fryer control screen, go to "Fryer Maintenance Screen", Select "Thermal Pump" ON to start running the Thermal pump to circulate the Thermofluid in the heating system. *(See Fig. C)*



Fig. B

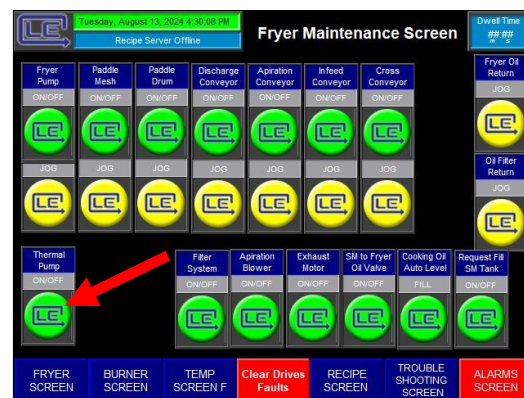


Fig. C

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER START UP PROCEDURES

9. Ensure the Boiler Main and Burner ball valves are fully open. The ball type valves are typically open when the handle is "in line" with the piping and valve body. *(See Fig. D and E)*



Fig. D



Fig. E

10. Check to ensure the pilot gas valve is in the "On" position. The "On" position indication is a red dot on the valve body, the "On" embossed on the valve handle needs to line up with the red dot on the valve body. To turn off the pilot gas valve, line up the "Off" embossing to the red dot. *(See Fig. F)*

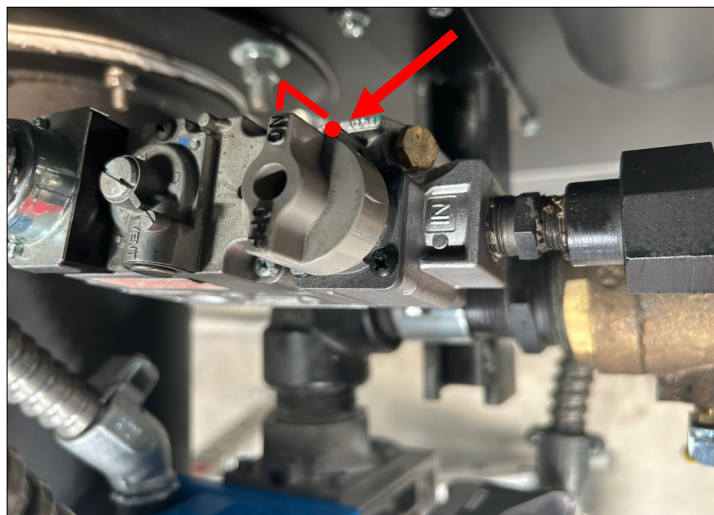


Fig. F

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER START UP PROCEDURES (continued)

11. From left to right, turn "On" the "Control Power", set the "Night Setback Switch" to "Off", Set the "Main Burner" Switch to "On", set the "Low Fire Hold" Switch to "Auto". (See Fig. G)

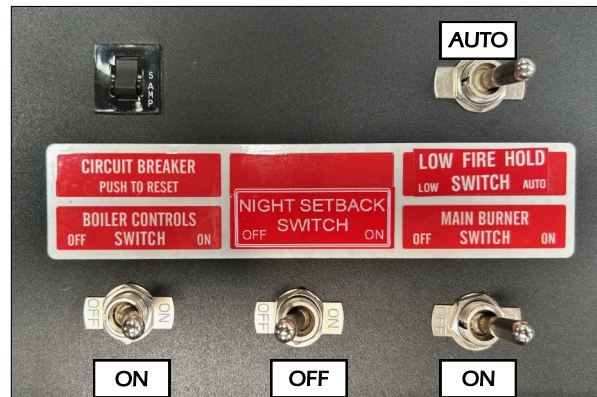
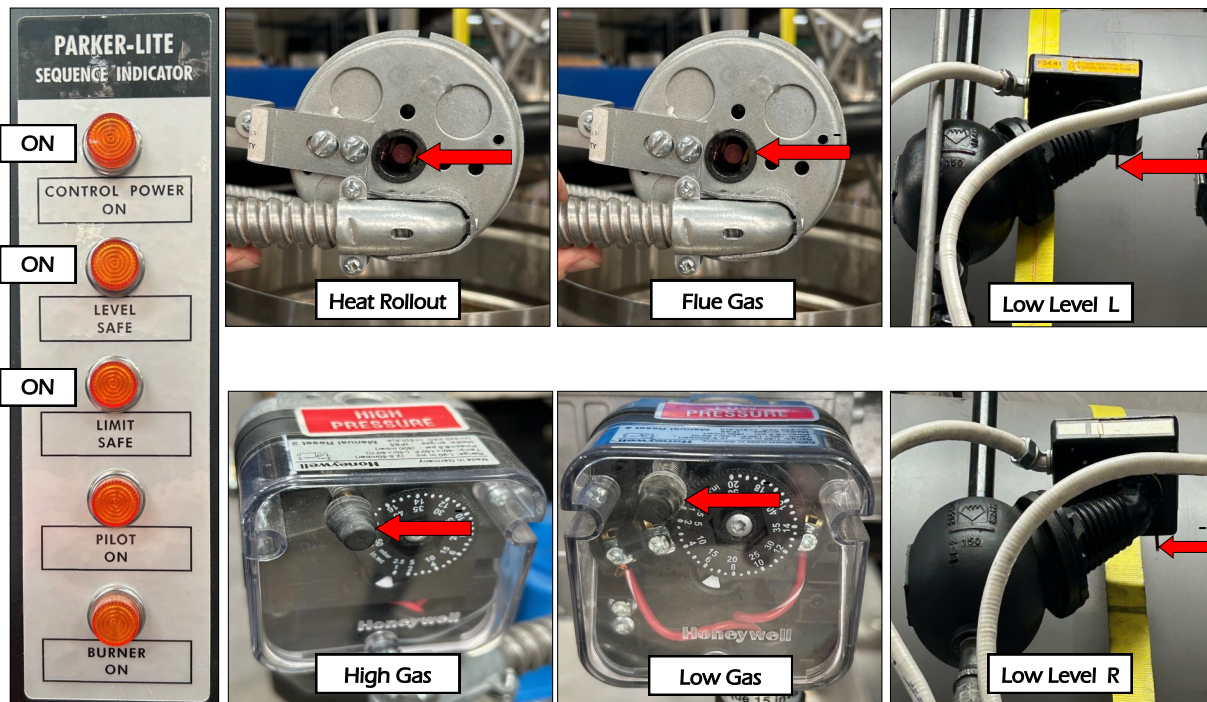


Fig. G

12. Check the status of the Sequence indicator pilot lights. The "Level safe" and "Limit safe" indicators should be illuminated. "Level Safe" not illuminated, check the expansion tank oil level and/or reset the low level switch. "Limit Safe" not illuminated, check and reset the High gas pressure switch, the low gas pressure switch, the flue gas spillage switch and the Heat rollout switch. If any of these switches are tripped, investigate the cause before resetting and operating the boilers. See pictures below AND NEXT PAGE for Pressure/level/safety switch identifications.



Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER START UP PROCEDURES (continued)

13. On top of the boiler control panel there are 5 toggle type switches *(See Fig. H)*
 - A. "Burner Controls Switch": Turns on the main power to the boiler. Does NOT initiate the flame or the burner processes.
 - B. "Night Setback Switch": Enables the "Night Setback" function which sets a lower temperature for night or hours of non operation.
 - C. "Main Burner Switch": Turns on the Fire Eye controller to start and monitor the burner process.
 - D. "Circuit Breaker": Protects the boiler control system from a high amperage fault. Investigate the reason for the circuit breaker trip. Push down on the toggle to reset.
 - E. "Low Fire Hold Switch": Enables the low fire hold function which maintains a low fire state in the burner until a certain pressure is made which then allows the burner to achieve a high fire state.
14. The front cover of the boiler control panel mounts 4 controls and 1 Status light array. *(See Fig. I)*
 - F. "Fire Eye" burner controller: Starts, monitors and maintains the burner pilot, burner and switch safety processes.
 - G. "Operating and modulating control": Sets and displays the Thermofluid temperature. Raise and lower the temperature of the Thermofluid by using the up arrow to raise, down arrow to lower then accepting the change by selecting the "Enter" key.
 - H. "Stack Temperature Control": Monitors the temperature of the boiler exhaust stack.
 - I. "Manual Reset High Limit": Monitors the high temperature state of the boiler, shuts off boiler in high temperature scenario, resets / restarts the boiler when the temperature drops below set point.
 - J. "Sequence Indicator": Illuminated Pilot Lights display the current status of the various systems. See next page for indicator definitions,



Fig. H

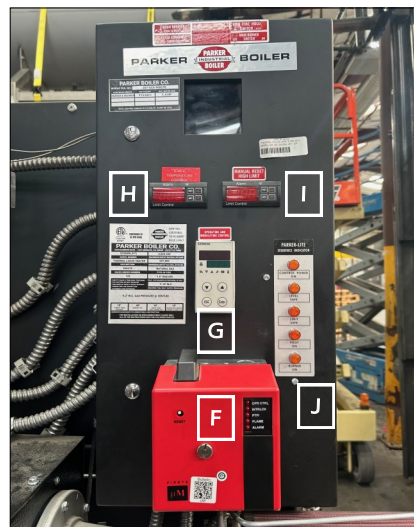


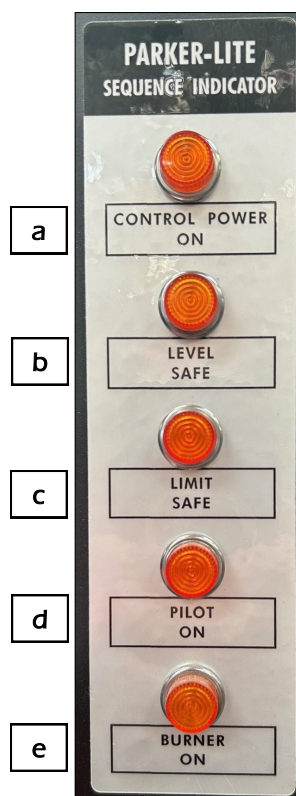
Fig. I

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER START UP PROCEDURES (continued)

- (a) "Control Power": When illuminated indicates the boiler control power is on. This is enabled by the toggle switch "Control Power On" which is located on top of electrical panel. See previous page.
- (b) "Level Safe": When illuminated indicates that the Thermofluid level in the expansion tank is sufficient to start up. If this does not illuminate check the Thermofluid level in the expansion tank, the level should be two inches above the indicating line on the low level switches mounted on the tank. Check the low level switch reset toggles. If a low level condition existed these toggles must be reset by pressing them in.
- (c) "Limit Safe": When illuminated indicates all of the pressure and safety switches are all in a safe condition. When not illuminated, this indicates a pressure or temperature switch was tripped and needs to be reset. Pressing in the reset button will restore the switch to a safe position.
- (d) "Pilot On": When illuminated indicates the pilot has been ignited and verified by the "Fire eye" burner controller.
- (e) "Burner On": When illuminated indicates the safety requirements have been met and it is safe to ignite the main burner. When not illuminated indicates the burner is not active possibly a fault or unsafe condition exists. Investigate the status of the "limit safe" and "level safe" sequence indicators.



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER START UP PROCEDURES (continued)

15. When the Burner is turned on with the toggle switch, the Fireeye controller will start up. The LED "OPR CTRL" and INTRLCK" will illuminate during the purge cycle.
16. The "PTFI" LED will illuminate during the Pilot ignition and test.
17. When the Pilot is proven, the "FLAME" LED will illuminate and the "PTFI" LED will now cease to be illuminated. The burner is not lit and ready to heat the Thermo fluid oil. (See Fig. J)
18. NOTE: If the "ALARM" LED is illuminated there is a fault in the sequence of events and the burner will not be enabled. See the side of the Fireeye controller for a detail of LED status for fault troubleshooting.

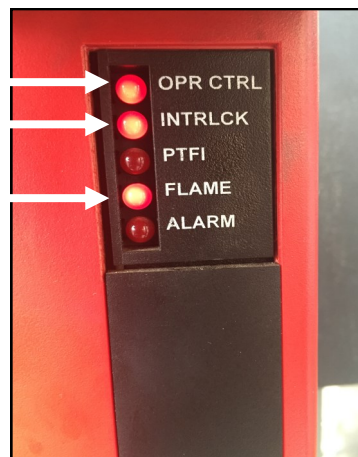


Fig. J

19. Verify the Set Points on both boiler Temperature controllers. Top display is the actual temperature, lower value is the Set Point. (See Fig. K and L)

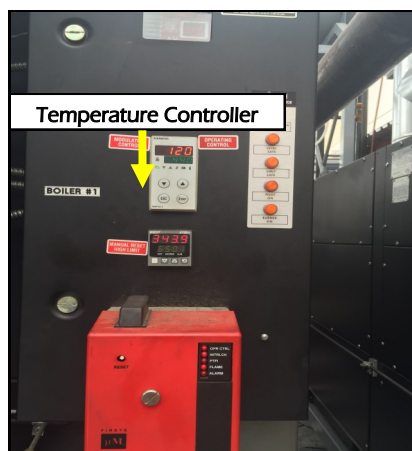


Fig. K

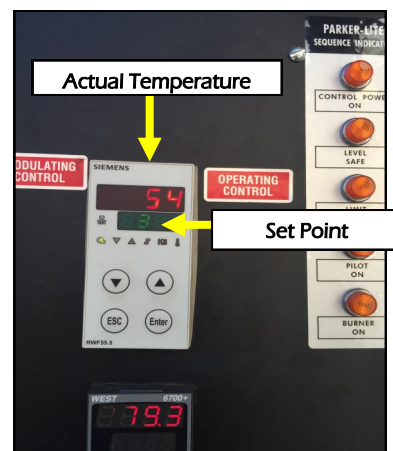


Fig. L

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER START UP PROCEDURES (continued)

20. If a Thermofluid temperature adjustment is needed, select the up or down arrow on the temperature controller to change the set point. Once the desired temperature is displayed in the set point window, select "Enter". If two boilers are being utilized, make this set change on both of them. (See fig. M and N)
21. At initial start up, set the Thermofluid temperature to 250 degrees Fahrenheit. Allow the system to heat up slowly.
22. Start at 250 degrees then allow to heat for 10 minutes, then raise to 350, let heat 10 minutes, then raise to 450 and allow to heat 10 minutes before going to production temperature.

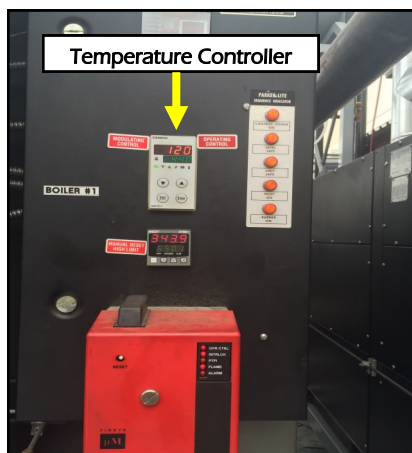


Fig. M

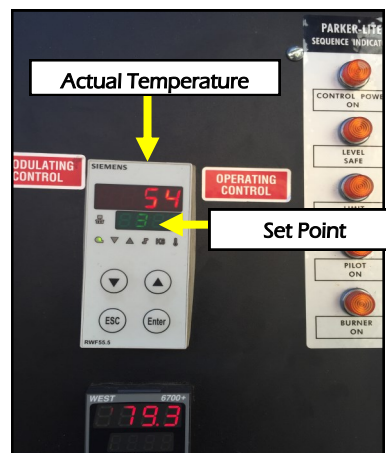
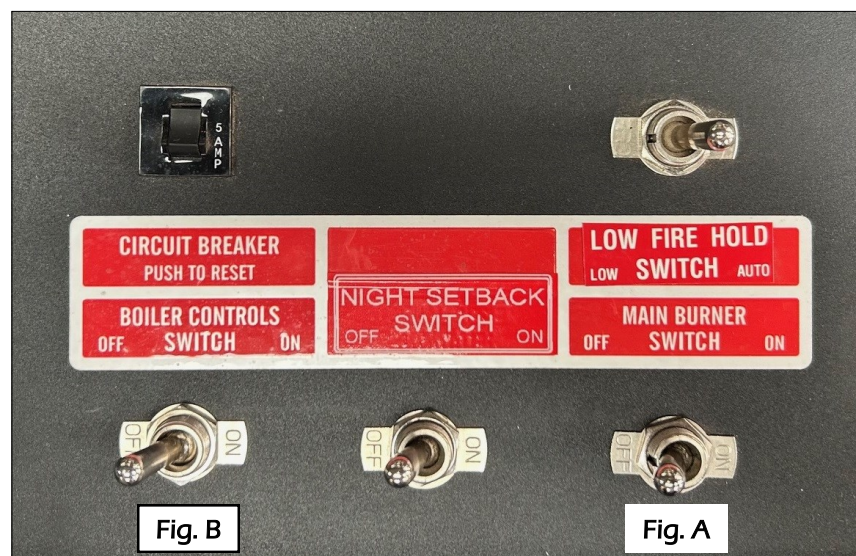


Fig. N

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PARKER BOILER SHUTDOWN PROCEDURES

1. Shut down Thermofluid boilers by selecting the toggle switch for Main Burner to the OFF position. Leave the Boiler circulation pumps and the Thermofluid pump running, allowing oil to circulate while cooling down. *(See Fig. A)*
2. Leave the Boiler controls On during the cool down.
3. When the Thermofluid has cooled down below 200°F, On the Boiler control panel, turn off boiler circulation pumps. DO NOT turn off Nitrogen or open the Nitrogen drain valve. DO NOT open any drain valves for the Thermofluid oil system.
4. When the Thermofluid is below 200 degrees Fahrenheit, turn off the Boiler Controls using the toggle switch. *(See Fig. B)*



BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS



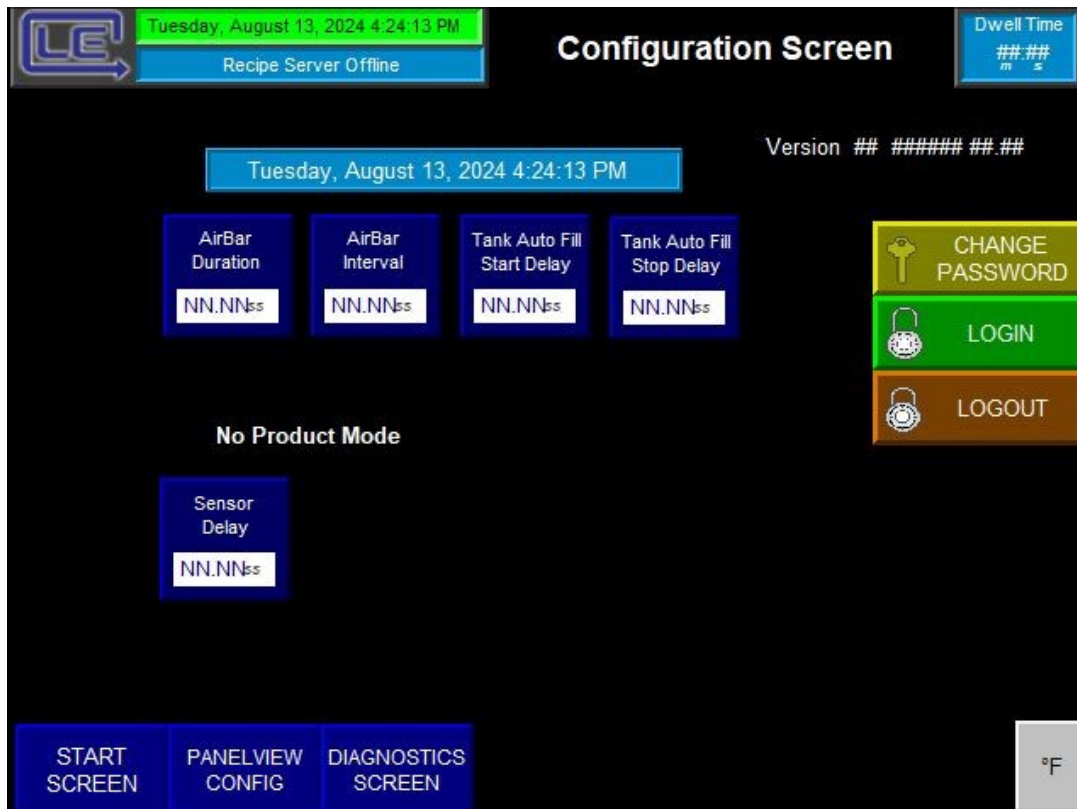
Start Up Screen– When the system is first powered up this is the screen that is displayed.

1. Fryer Screen: This setting allows you to Access the Fryer Screen.
2. Configuration Screen: This Configuration Screen allows you to access setup parameters, however, these screens should only be accessed by Lawrence Equipment Technical personnel.
3. Spanish/English: Allows the user to change screen language.

Note: Some Screens are password protected. Only authorized personnel and Lawrence Equipment should access these screens. Before accessing password protected screens consult Lawrence Equipment, as improper adjustment of any of the values could cause extensive damage to the machine.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



Configuration Screen- This screen allows you to enter the date (year, month and year), and time (hour, minute and seconds).

1. Change Password: This setting allows the user to change his/her password. The user must first login using their existing username and password.
2. Login: This setting will allow you to login to your account using your user name and password.
3. Logout: This setting allows the user to immediately logout of their account. This is recommended after an adjustment is made so there are no unauthorized adjustments made. **Note:** Session will automatically logout after 5 minutes of inactivity.
4. Air Bar Duration: This numeric entry adjusts how long the Air Bar in the Oil Filter box is activated. A longer value will have the air bar blowing air onto the filter drum longer. A setting of 20 seconds means the air bar will blow air onto the filter drum for 20 seconds. *(Setting 0 ~ 60 sec.)*
5. Air Bar Interval: This numeric entry adjusts the time between Oil Filter box air bar is activations. A longer interval means more time between air activations/Duration. A setting of 20 seconds means a 20 second gap between air bar blows. *(Setting 0 ~ 300 sec.)*
6. Tank Auto Fill Start Delay: This numeric entry adjusts the time that the tank Auto Fill function is activated. This delay starts when the tank low level float is activated by a low fluid level. *(Setting 0 ~ 10min)*

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

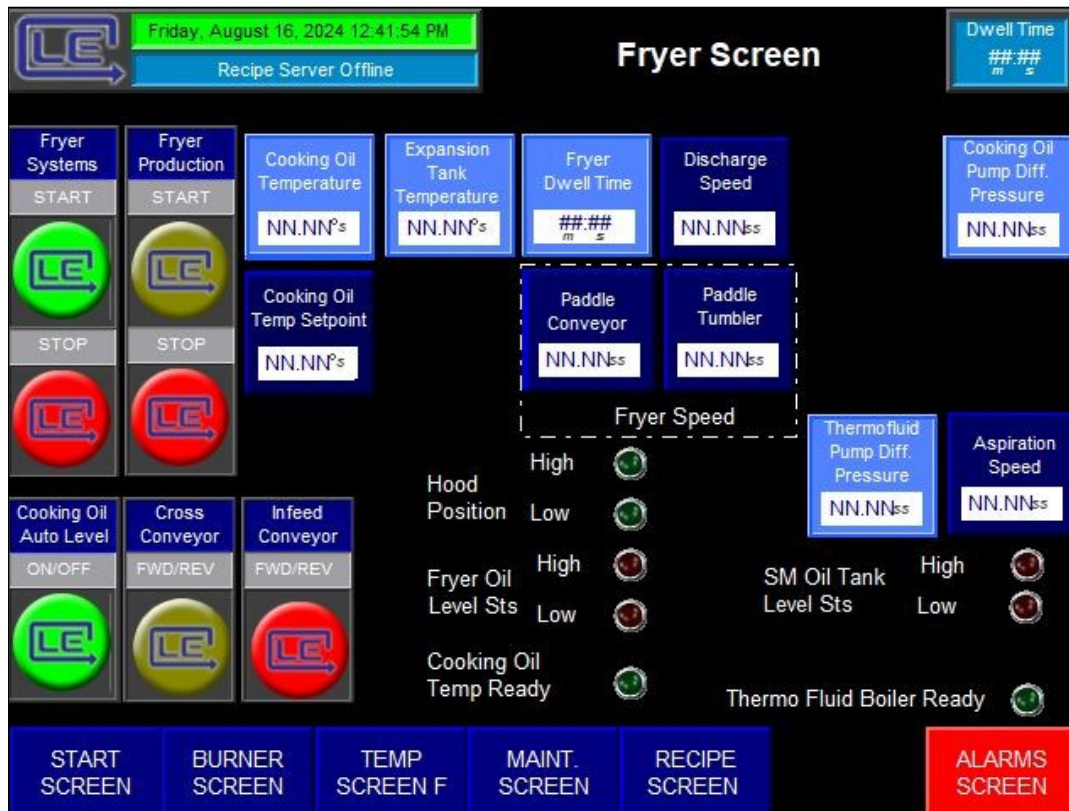
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

7. Tank Auto Fill Stop Delay: This numeric display adjusts the time for the Auto Fill function to stop. Increasing the value allows for more time before the tank fills stops. In example... the Tank auto fill start delays for 5 minutes, when the delay is done the tank fill function starts and the tank auto fill stop delay is activated. With a setting of 5 minutes, the tank will continue to fill for 5 minutes then stop.
8. Product Sensor: This numeric entry adjusts the product detect time. The time delay starts when the product sensor does not detect chips on the fryer infeed conveyor. A setting of 5 seconds means when there are no chips going into fryer for 5 seconds before the system goes into no product mode. *(Setting 0 ~ 10 sec.)*
9. Start Screen: This setting allows the user to access the Start Screen.
10. Panel View Configuration Screen: This setting allows the user to access the Panel View Configuration Screen.
11. Diagnostics Screen: Allows the user to access the Diagnostics Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



Fryer Screen

1. Fryer System (Start/Stop): This function will turn on all the systems of the fryer, including: Infeed Conveyor, Paddle conveyor, Fryer Discharge conveyor, Cooking Oil Pump, Aspiration conveyor, Aspiration blower, Aspiration blower, Oil Collector Discharge, Filter System and Cooking Oil AutoFill function. This function will not turn on the **Heat Exchanger** systems. (*Please Refer to Heat Exchanger Start-up*).
2. Fryer Production (Start/Stop): This selection will place the system in production mode. This places the cooking oil temperature to the operational run temperature.
3. Cooking Oil Auto Level (On/Off): Select on or off to control the cooking oil level in the kettle automatically. This will allow the system to feed oil very slowly from the storage tank as needed.
4. Cooking Oil Temperature Set point: This is the desired cooking oil temperature. To change it, select the numeric display and input the desired temperature. (*Setting 32°F ~ 450°F*)
5. Expansion Tank Temperature: This indicator displays the current temperature of the Thermo fluid in the Expansion tank. This temperature should not rise above 195 degrees Fahrenheit.
6. Fryer Dwell Time: This indicator displays the current fryer dwell time. This dwell is the actual time the chips are in the fryer from infeed to discharge.

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

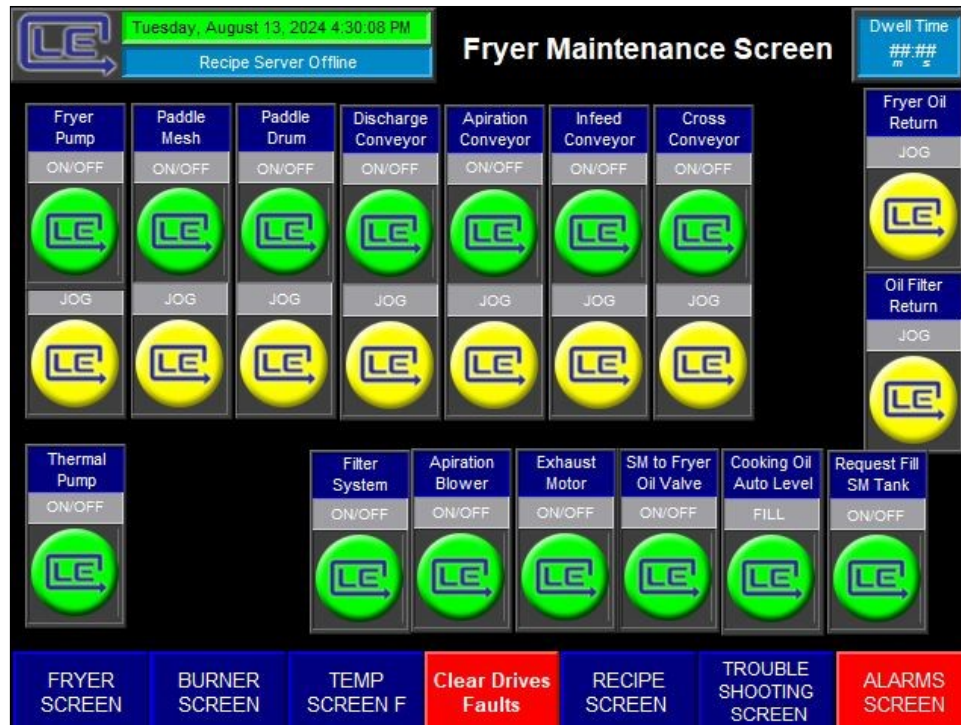
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

7. Discharge Speed: This indicator is used to increase or decrease the discharge belt speed. This is indicated in percent of potential run speed. *(Setting 10 ~ 100%)*
8. Paddle Conveyor: This setting adjusts the speed of the Fryer Paddle Conveyor. Increasing this value will increase the speed, decreasing the value lowers the speed. *(Setting 5 ~ 120%)*
9. Paddle Tumbler: This setting adjusts the speed of the fryer paddle conveyor. Increasing this value will increase the speed, decreasing the value lowers the speed. *(Setting 5 ~ 120%)*
10. Cooking Oil Pump Diff. Pressure: Displays pressure differential between the inlet and outlet of the heat exchanger. High input pressure vs. low output pressure can be signs of a obstruction in pipe.
11. Thermo fluid Pump Diff Pressure: This indicator displays the current differential pressure of the Thermo fluid. This value is the pressure difference from the inlet of the thermo fluid heat exchanger to the outlet of the thermo fluid heat exchanger.
12. Aspiration Speed: This setting allows the user to adjust the speed of the Aspiration conveyor. Increasing the speed will lower the total overlap of the product, decreasing the speed will slow the belt down increasing the overlap. *(Setting 10 ~ 100%)*
13. Hood Low Position: Green illumination Indicates the Hood is in the Low Position.
14. Hood High Position: Green illumination indicates the hood is the high position.
15. Fryer Oil Low Level Sts (Status): Illumination indicates that the oil level in the Fryer kettle is low.
16. Fryer Oil High Level Sts Status: Illumination indicates that the oil level in the Fryer kettle is High.
17. Cooking Oil Temperature Ready: Illumination indicates that the Cooking Oil Temperature is ready.
18. Tank Oil Low Level Sts (Status): Indicates that the storage tank oil level is low.
19. Tank Oil High Level Sts (Status): Indicates the storage tank oil level is high.
20. Thermo Fluid Boiler: Green illumination indicates the Thermo Fluid boilers are running.
21. Start Screen: This setting allows access to the Start Screen.
22. Temperature Screen: This setting allows access to the Temperature screen.
23. Maintenance Screen: This setting allows access to the Maintenance Screen.
24. Recipe Screen: This setting allows access to the Recipe Screen.
25. Alarm Screen: This setting allows access to the Alarm Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



Fryer Maintenance Screen

1. Fryer Pump On/Off: This setting turns the fryer cooking oil pump On and Off.
2. Fryer Pump (Jog): This setting will jog the Fryer oil pump. The pump will run as long as the setting is maintained.
3. Paddle Mesh On/Off: This setting turns On and Off the Fryer Paddle mesh belt conveyor.
4. Paddle Mesh (Jog): This setting will run the mesh paddle conveyor at low speed as long as the setting is maintained.
5. Paddle Drum Off/On: Turns On/Off the Paddle Drum.
6. Paddle Drum (Jog): Momentarily Jogs the Paddle Drum.
7. Discharge Conveyor On/Off: Turns On and Off the Chip Fryer discharge conveyor.
8. Discharge Conveyor (Jog): This setting runs the discharge conveyor as long as the setting is maintained.
9. Aspiration Conveyor On/Off: Turns On/Off the Aspiration Conveyor.
10. Aspiration Conveyor (Jog): Momentarily Jogs the Aspiration Conveyor.
11. Infeed Conveyor On/Off: Turns On/Off the Infeed Conveyor feeding chips into the fryer kettle.
12. Infeed Conveyor (Jog): Momentarily Jogs the Infeed Conveyor.

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

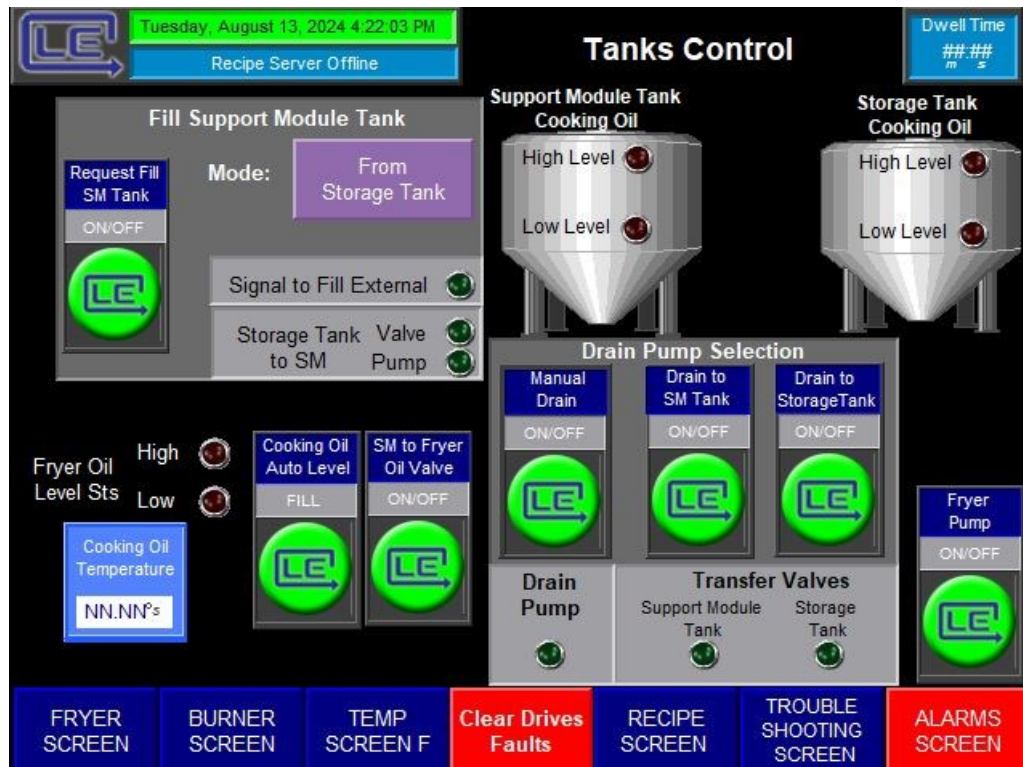
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

13. Cross Conveyor On/Off: Turns On/Off the Cross Conveyor after the Aspiration Conveyor.
14. Cross Conveyor (Jog): Momentarily Jogs the Cross Conveyor.
15. Fryer Oil Return (Jog): This selection temporarily jogs the fryer oil return. The pump will only stay on as long as the setting is maintained, releasing it stops the pump.
16. Oil Filter Return (Jog): This selection temporarily jogs the Fryer oil filter return pump. The pump will only stay On as long as the setting is maintained, releasing it stops the pump.
17. Thermal Pump On/Off: Turns On/Off the Thermal Fluid Pump on the Boiler module.
18. Filter System On/Off: This setting turns the fryer filter On and Off to remove debris in the Cooking Oil. (This drum filter is located under the heat exchanger).
19. Aspiration Blower On/Off: Turns On/Off the Aspiration Blower.
20. Exhaust Motor On/Off: This setting turns On and Off the fryer exhaust blower. **Note:** This Exhaust motor evacuates the fumes/air from the fryer kettle through the Quick exhaust system.
21. SM to Fryer Oil Valve On/Off: This button will open, if on, the fryer cooking oil valve. Make sure that you use this function for maintenance purposes only, **when this function is active the fryer cooking oil valve will remain open regardless of the oil level in the fryer.**
22. Cooking Oil Auto Level (Fill): This setting will turn on the Auto Fill function, this means that the fryer will control the cooking oil level at it's optimum operational level automatically.
23. Request Fill SM Tank: On/Off: This setting enables the Support Module tank oil request. This will send signal to fresh oil feed pump.
24. Fryer Screen: This setting allows access to the Fryer Screen.
25. Temperature Screen: This setting allows you to access the Temperature Screen.
26. Clear Drive Faults: This setting will clear and reset any drive faults. Any active faults will not clear.
27. Recipe Screen: This setting allows you to access the Recipe Screen.
28. Troubleshooting Screen: This setting allows you to access the Troubleshooting screen.
29. Alarm Screen: This setting allows you to access the Alarms Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



Tanks Control

1. Request Fill SM Tank On/Off: Turns On/Off the request signal to fill the Support module tank.
2. Mode: From Storage Tank: Momentarily selecting this scrolls from Storage tank to External tank.
3. Signal to Fill External: Illuminates when the signal is sent to fill oil in the external tank.
4. Storage Tank to SM Valve: Illuminates when the Storage tank to Support Module valve is open.
5. Storage Tank to SM Pump: Illuminates when the Storage Tank to Support Module pump is running.
6. Fryer Oil Level STS High: Illuminates when the Fryer kettle oil level is high.
7. Fryer Oil Level STS Low: Illuminates when the Fryer kettle oil level is low.
8. Cooking Oil Auto Level (Fill): This selection enables the cooking oil auto level function.
9. SM to Fryer Oil Valve On/Off: This selection turns On/Off the Support Module to Fryer valve.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

10. Support Module Tank Cooking Oil:

- * High Level: Illuminates when the cooking oil reaches the high level limit.
- * Lower Level: Illuminates when the cooking oil reaches the low level limit.

11. Support Tank Cooking Oil:

- * High Level: Illuminates when the cooking oil reaches the high level limit.
- * Lower Level: Illuminates when the cooking oil reaches the low level limit.

12. Manual Drain On/Off: Turns On/Off the Drain pump

13. Drain Pump: Illuminates when the Drain pump is running.

14. Drain to SM Tank On/Off: Turns On/Off the Drain to Support Module Tank valve.

15. Drain to Storage Tank On/Off: Turns On/Off the Drain to Storage Tank valve.

16. Fryer Pump On/Off: Turns On/Off the Fryer pump.

17. Transfer Valves:

- * Support Module Tank: Illuminates when the support module tank valve is activated.
- * Storage Tank: Illuminates when the storage tank valve is activated.

18. Fryer Screen: This setting allow you to access the Fryer Screen

19. Burner Screen: This setting allows the user to access the Burner Screen.

20. Temperature Screen: This setting allows you to access the Temperature Screen.

21. Clear Drives Faults: This setting will clear and reset any drive faults. Any active faults will not clear.

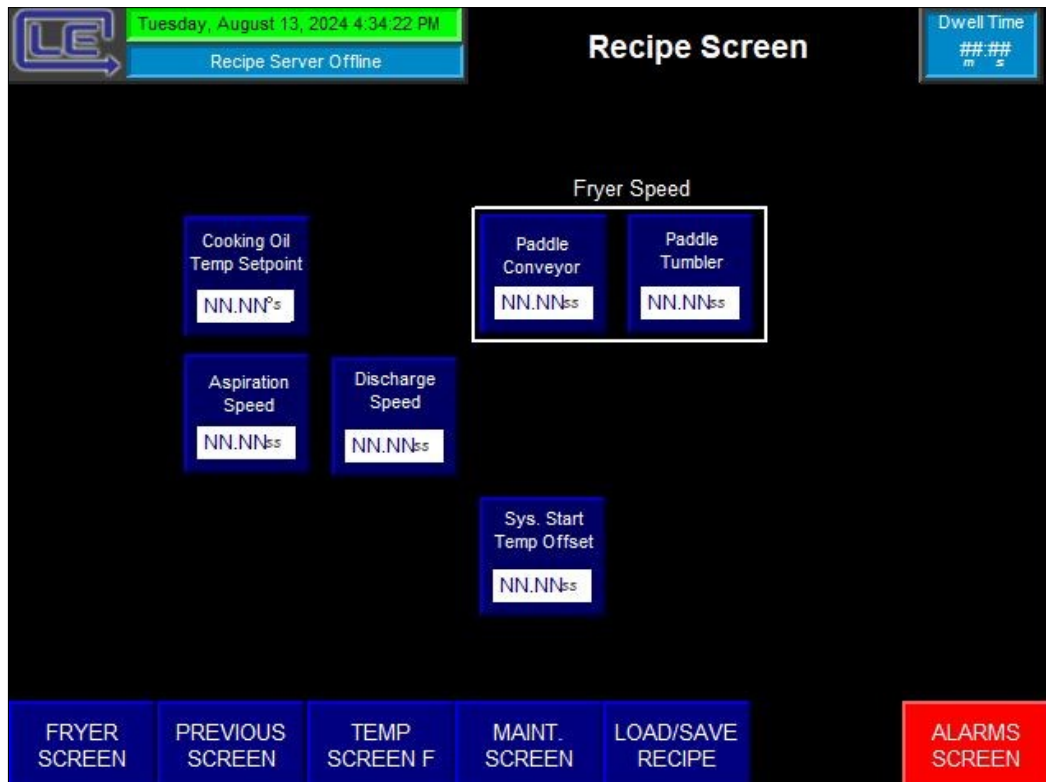
22. Recipe Screen: This setting allows you to access the Recipe Screen.

23. Troubleshooting Screen: This setting allows you to access the Troubleshooting screen.

24. Alarms Screen: This setting allows you to access the Alarms Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



Recipe Screen

1. Cooking Oil Temperature Set point: This setting allows you to control the temperature of the cooking oil. *(Setting 32 °F ~ 450°F)*
2. Aspiration Speed: This setting adjusts the speed of the Aspiration Conveyor belt. *(Setting 10 ~ 100%)*
3. Discharge Speed: This setting allows the user to adjust the speed of the fryer discharge. Increasing the speed will increase the product flow out of the fryer. *(Setting 10% ~ 100%)*
4. Fryer Speed: This setting allows the user to control the frying time in seconds. *(Setting 5% ~ 120%)*
5. Paddle Conveyor: This setting adjusts the speed of the Paddle Conveyor. *(Setting 5 ~ 120%)*
6. Paddle Tumbler: This setting adjusts the speed of the Paddle Tumbler. *(Setting 5 ~ 120%)*
7. System Start Temperature Offset: This sets the temperature slightly lower than the normal cook temperature for startup. *(Setting 0 ~ 60°F)*
8. Fryer Screen: This setting allows you to access the Fryer Screen.

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

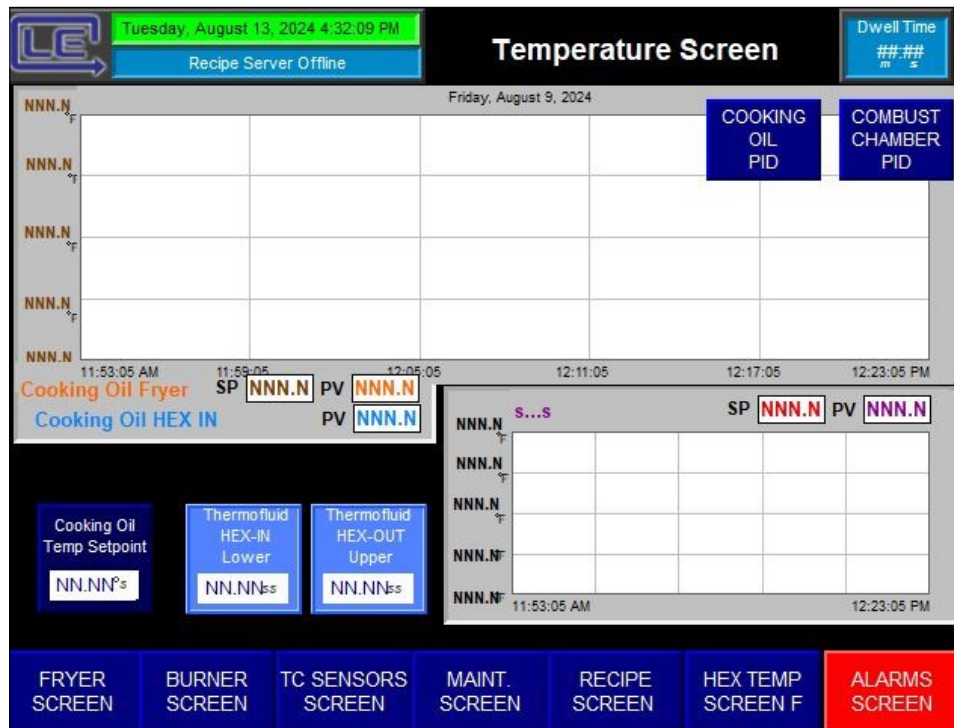
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

- 9. Previous Screen: This setting allows you to access the Previous Screen.
- 10. Temperature Screen: This setting allows you to access the Temperature Screen.
- 11. Maintenance Screen: This setting allows you to access the Maintenance Screen.
- 12. Load/Save Recipe: This setting allows the user to access the Load/Save Recipe Screen.
- 13. Alarm Screen: This setting allows you to access the Alarm Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

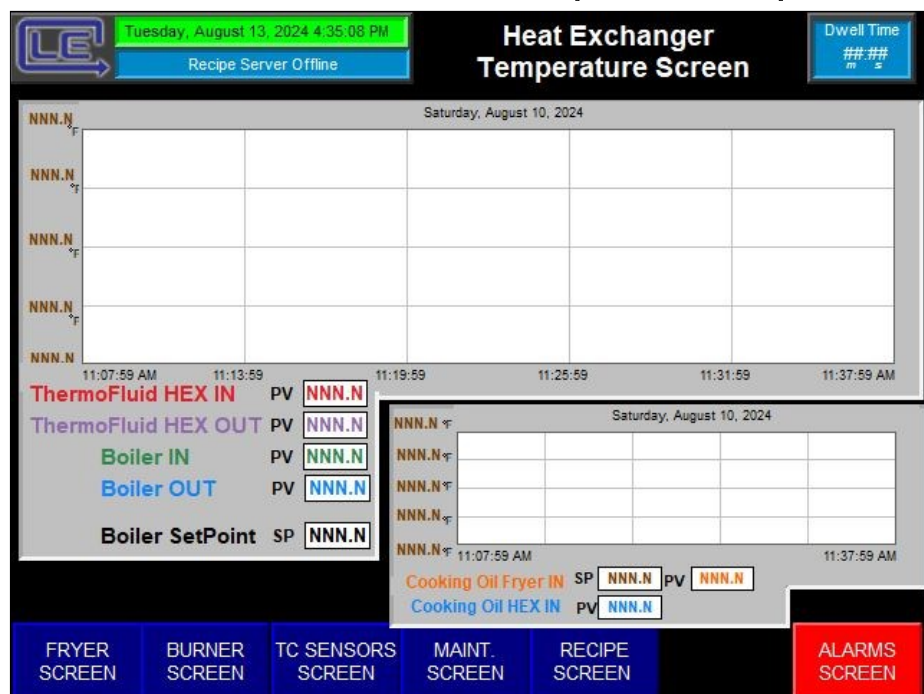


Temperature Screen

1. Cooking Oil PID: Access to the adjustments to the PID program. Technicians only.
2. Combustion Chamber PID: Access to the adjustments to the PID program. Technicians only.
3. Cooking Oil Temperature Set point: This setting allows you to control the temperature set point for the cooking oil. This setting should be set at the optimal temperature for the product being run.
(Setting 32 ~ 450 °F)
4. Thermo fluid Upper Out TC4: This display is the current Thermo fluid temperature measured at the output of the Thermo fluid heat exchanger.
5. Thermo fluid Lower in TC3: This display is the current Thermo fluid temperature measure at the input of the Thermo fluid heat exchanger.
6. Cooking Oil Process Value (PV): This indicator displays the current temperature of the cooking oil
7. Cooking Oil HEX IN (PV): This indicator displays the current temperature of the cooking oil going into the Heat Exchanger.
8. Expansion Tank (PV): This Display is the current temperature of the Thermo Fluid Expansion tank.
9. Fryer Screen: This setting allows the user to access the Fryer Screen.
10. Previous Screen: This setting allows the user to access the Previous Screen.
11. Recipe Screen: This setting allows the user to access the Recipe Screen.
12. Alarm Screen: This setting allows the user to access the Alarm Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



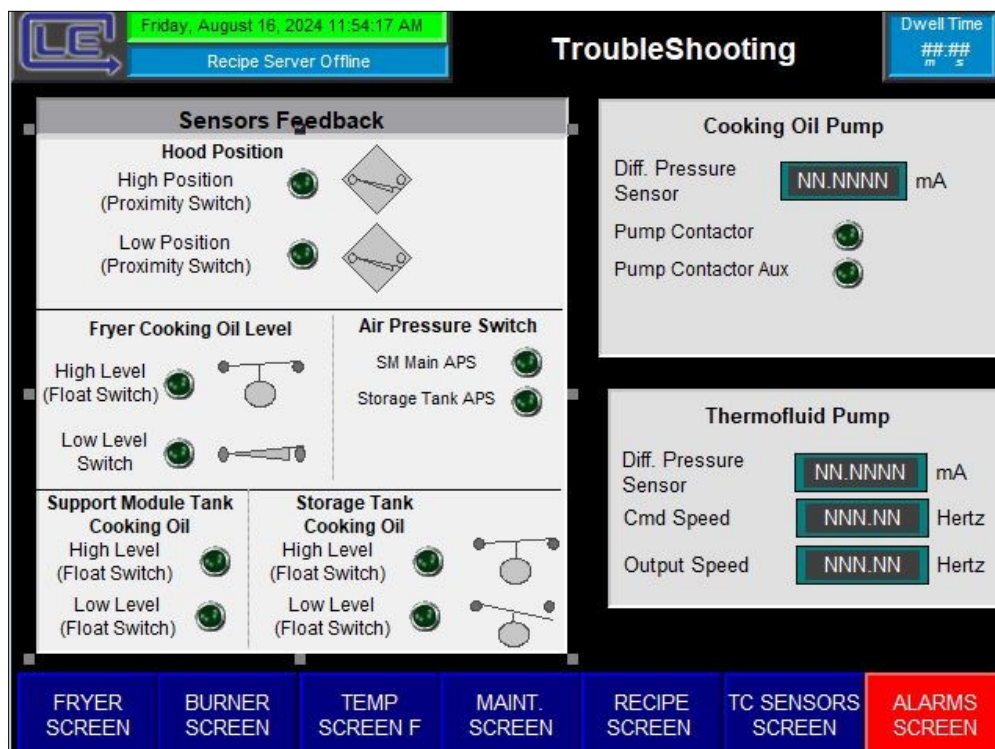
Heat Exchanger Temperature Screen

1. ThermoFluid HEX IN (PV): Displays the temperature of the thermofluid entering the Heat Exchanger.
2. ThermoFluid HEX OUT (PV): Displays the temperature of the thermofluid exiting the Heat Exchanger.
3. Boiler IN (PV): Displays the temperature of the thermofluid entering the Boilers.
4. Boiler OUT (PV): Displays the temperature of the thermofluid exiting the Boilers.
5. Boiler Setpoint (SP): This is the temperature you want the thermofluid oil to get to.
6. Cooking Oil Fryer IN (SP): This is the temperature you want the cooking oil to get to for production.
7. Cooking Oil Fryer IN (PV): Displays the temperature of the cooking oil entering the fryer.
8. Cooking Oil HEX IN (PV): Displays the temperature of the cooking oil entering the heat exchanger.
9. Fryer Screen: This setting allows the user to access the Fryer Screen.
10. Burner Screen: This setting allows the user to access the Burner Screen.
11. TP Sensors Screen: This setting allows the user to access the TP Sensors screen.
12. Maintenance Screen: This setting allows the user to access the Maintenance Screen.
13. Recipe Screen: This setting allows the user to access the Recipe Screen.
14. Alarm Screen: This setting allows the user to access the Alarm Screen.

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



Troubleshooting

1. Hood Position High Position (Proximity Switch): This indicates the fryer kettle hood is in the up (high) position. The fryer conveyors will NOT operate when the hood is in the up position.
2. Hood Position Low Position (Proximity Switch): This indicates the fryer kettle hood is in the down (low) position. The fryer kettle hood MUST be in the low position in order to operate the paddle conveyors.
3. Fryer Cooking Oil High Level (Float Switch): This indicates that the fryer kettle oil level high switch is activated. High oil level detected. Immediately check the reason for the kettle oil high level state. Potential for kettle overflow is very high at this time. The high level switch will activate the drain pump to lower the cooking oil level in the kettle sending excess cooking oil to the storage tank.
4. Fryer Cooking Oil Low Level: This indicates that the fryer kettle oil level low switch is activated. Low oil level detected. Check status of the auto fill level probe and the cooking oil level in the support module tank.
5. Air Pressure Switch (SM Main APS): This indicator will illuminate when the air pressure switch on the Support Module is being activated. When the indicator is **NOT** illuminated it is signaling that air pressure has been lost. Verify the air supply hose is connected, verify the air manifold manual valve is open, verify the safety system is energized, verify that compressed air is being supplied by plant compressor.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

6. Air Pressure Switch (Storage Tank APS): This indicator will illuminate when the air pressure switch on Storage tank air manifold is activated. When the indicator is **NOT** illuminated it is signaling that air pressure has been lost. Verify the air supply hose is connected, verify the air manifold manual valve is open, verify the safety system is energized, verify that compressed air is being supplied by plant compressor
7. Support Module Tank Cooking Oil (High Level Float Switch): This indicator will illuminate when the cooking oil in the support module tank is in a high level state. **DO NOT** continue filling the tank when this indicator is illuminated.
8. Support Module Tank Cooking Oil (Low Level Float Switch): This indicator will illuminate when the cooking oil in the support module tank is in a low level state. If the tank is being used as a “make up” during operations it is imperative to add cooking oil or review the fresh cooking oil supply availability. Failure to maintain a fresh supply of cooking oil will cause a low level state in the fryer kettle.
9. Storage Tank Cooking Oil High Level (Float Switch): This indicator will illuminate when the cooking oil in the storage tank is at a high level. Do not continue to fill the tank when this indicator is illuminated.
10. Storage Tank Cooking Oil Low Level (Float Switch): This indicator will illuminate when the cooking oil in the storage tank is at a low level. If the storage tank is supplying fresh cooking oil to the support module tank investigate the reason that the storage tank is low and rectify the problem.
11. Cooking Oil Pump (Diff. Pressure Sensor): This display indicates the cooking oil pressure difference between the input and output of the heat exchanger. This is used to not only to confirm that the pump is running but also too high of an input pressure and a low output pressure could be a sign of a blockage in the heat exchanger. Any large difference in the pressures needs to be investigated. Verify any valves in the circuit are open or closed according to the state of the fryer and the operation that is currently being performed.
12. Cooking Oil Pump (Pump Contactor): This indicator is illuminated when the cooking oil pump contactor is activated. The cooking oil pump should be running at this time. If the indicator is illuminated, the contactor closed yet the pump not running, verify the presence of three phase power supplying the contactor. Verify all manual motor disconnects are currently turned on.
13. Cooking Oil Pump (Pump Contactor Aux): This indicator will illuminate when the cooking oil pump contactor is energized and closed. The auxiliary contact is mechanically connected to the contactor. If the pump is running (contactor closed) but no indicator illumination, check the status of the aux contact on the contactor. Verify control voltage on both sides of the aux contacts.
14. ThermoFluid Pump (Diff. Pressure Sensor): This display indicates the ThermoFluid pressure difference between the input and output of the heat exchanger. This is used to not only to confirm that the pump is running but also too high of an input pressure and a low output pressure could be a sign of a blockage in the heat exchanger. Any large difference in the pressures needs to be investigated. Verify any valves in the circuit are open or closed according to the state of the fryer and the operation that is currently being performed.

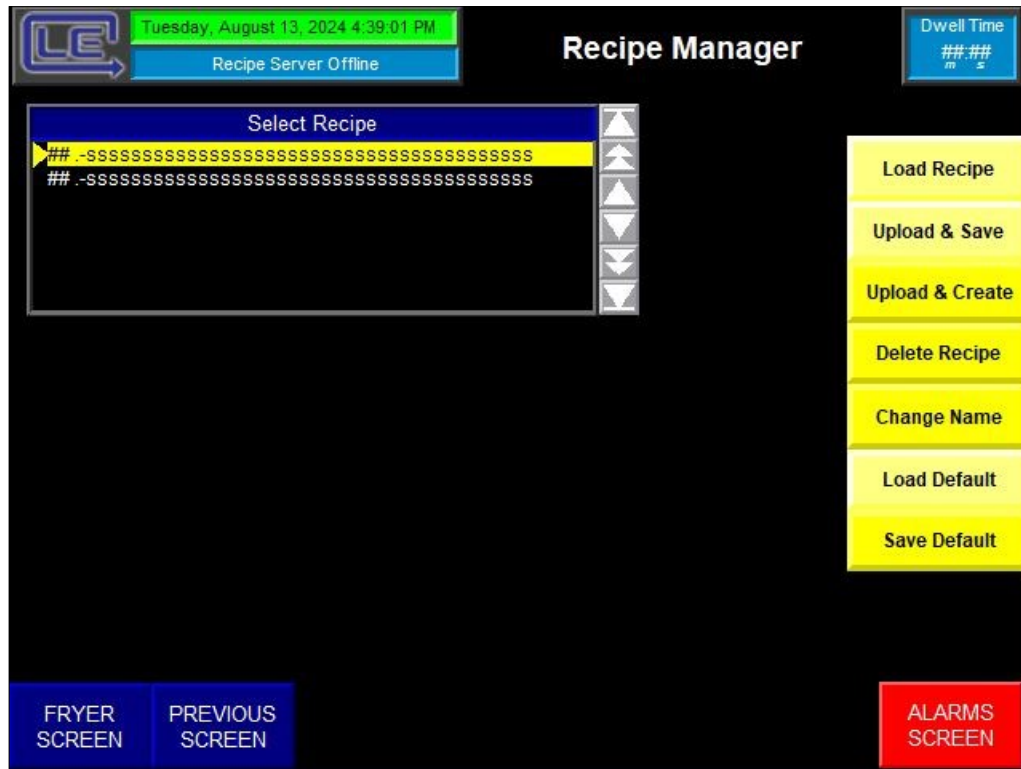
BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

15. Thermofluid Pump (Cmd Speed): This numeric display is the current command speed being sent to the Thermofluid pump frequency drive from the PLC. This speed is automatically monitored and set by the Programmable Logic Controller in the fryer AND boiler control panels.
16. Thermofluid Pump (Output Speed): This numeric display is the current output speed the frequency drive is sending to the Thermofluid pump motor. This speed is in reference to the command speed being sent from the PLC.
17. Fryer Screen: This setting allows the user to access the Fryer screen.
18. Burner Screen: This setting allows the user to access the Burner controls screen.
19. Temperature Screen F: This setting allows the user to access the Temperature screen.
20. Maintenance Screen: This setting allows the user to access the Maintenance screen.
21. Recipe Screen: This setting allows the user to access the Recipe Screen.
22. TC Sensors Screen: This setting allows the user to access the Thermocouple Sensors screen.
23. Alarms Screen: This setting allows the user to access the Alarms Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)

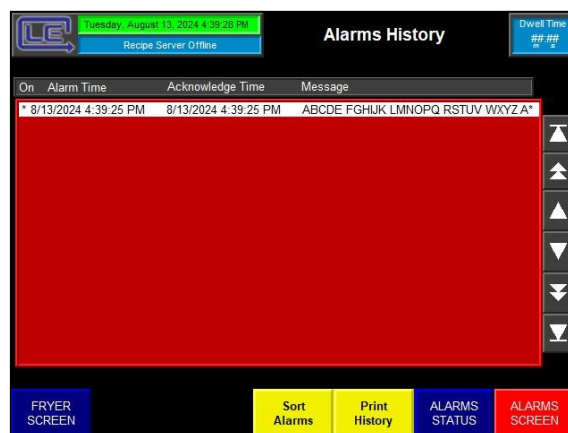
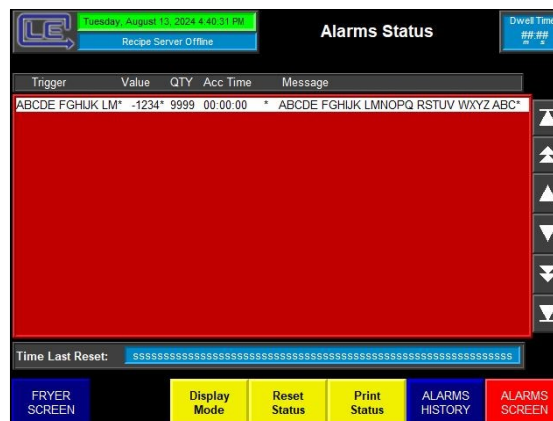


Recipe Manager

1. Load Recipe: This setting allows the user to load the selected recipe.
2. Upload & Save: This setting allows the user to load the current settings and save the selected recipe.
3. Upload & Create: This setting allows the user to upload the current settings for the new recipe.
4. Delete Recipe: This setting allows the user to delete a recipe.
5. Change Name: This setting allows the user to change recipe name.
6. Load Default: **Note:** This setting should only be accessed by Lawrence Equipment.
7. Save Default: **Note:** This setting should only be accessed by Lawrence Equipment.
8. Fryer Screen: This setting allows the user to access the Fryer Screen.
9. Previous Screen: This setting allows the user to access the Previous Screen.
10. Alarm Screen: This setting allows the user to access the Alarm Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

PANEL DESCRIPTIONS (continued)



Alarms Screen: These screens will appear if any faulted conditions or safety system violations are displayed. These are different levels of alarms. Some stop the production of the line and prevent it from re-starting. Other non-critical alarms do not prevent it from starting but will remain on the screen until the condition is cleared. The alarm setting will flash red when there is an alarm. Alarms should always be addressed as soon as possible to keep the system running to its optimum of safety and performance.

Alarm Status: Displays alarm status setting of all user configurable alarm.

Alarm History: Displays a list of all the alarms that have occurred since last power up.

1. Main Menu Screen: Allows you to access the Main Menu Screen.
2. Previous Screen: Allows you to access the Previous screen.
3. Silence Alarms: Allows you to silence an alarm but will not acknowledge the alarm.
4. Acknowledge Alarms: This setting will allow you to clear any alarms after the condition that created the alarm has been repaired or addressed.
5. Ack All Alarms: Allows you to acknowledge all alarms have been repaired or addressed.
6. Alarm History: Allows you to access the Alarm History Screen.
7. Alarm Status: Allows you to access the Alarm Status Screen.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

ALARM DESCRIPTIONS

Possible Alarm Messages	Description/Action
Fryer Paddle Conveyor AC Drive Fault	This indicates a fault condition in the AC inverter located inside fryer electrical panel. For a description of fault codes refer to the AB PowerFlex AC Drive manual supplied with the machine.
MMP 10 Cook Oil Pump Overload	This indicates that the cooking oil pump motor has tripped the thermal overload device, located in the fryer power electrical panel. To clear this alarm reset the overload. Investigate reason for overload, check FLA of motor vs. metered amps.
Aspiration Conveyor AC Drive Fault	This indicates a fault condition in the AC inverter located inside the fryer electrical panel. For a description of fault codes refer to the AB PowerFlex AC Drive manual supplied with the machine.
MMP5 Aspiration Blower Overload	This indicates that the aspiration blower motor has tripped the thermal overload device, located in the fryer power electrical panel. To clear this alarm reset the overload. Investigate reason for overload, check FLA of motor vs. metered amps.
Fryer Exhaust Blower Fault	This indicates that there is a fault condition on the exhaust blower motor. Check Overloads and motor protectors in fryer electrical panel.
MMP11 Fryer Hood Motor Overload	This indicates that the fryer hood motor has tripped the thermal overload device, located in the fryer power electrical panel. Check the motor protectors in the fryer electrical panel. Investigate reason for overload. Check FLA vs. metered amps.
MMP1 Fryer Drain Pump Motor Overload	This indicates that the fryer pump motor has tripped the thermal overload device, located in the fryer electrical panel. To clear this alarm reset the overload. Investigate the reason for the overload. Check motor FLA vs. metered amps.
MMP2 Cook Oil Filter Belt Overload	This indicates that the cooking oil filter motor has tripped the thermal overload device, located in the fryer power electrical panel. To clear this alarm reset the overload. Investigate the reason for the overload. Check the FLA vs. metered amps.
MMP3 Cook Oil Filter Drum Overload	This indicates that the cooking oil filter drum motor has tripped the thermal overload device, located in the fryer power electrical panel. To clear this alarm reset the overload. Investigate reason for the overload. Check the FLA vs. metered amps.
MMP6 Weight Conveyor Motor Overload	This indicates that the weight conveyor motor has tripped the thermal overload device, located in the fryer power electrical panel. To clear this alarm reset the overload. Investigate reason for the overload. Check the FLA vs. metered amps.
Cooking Oil Low Pressure Switch Fault	This indicates that there is not enough cooking oil pressure in the system, check for oil level and proper functioning of the cooking oil pump or check for a faulty pressure switch.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

ALARM DESCRIPTIONS (continued)

Possible Alarm Messages	Description/Action
Cooking Oil High Pressure Switch Fault	This indicates that there is too much cooking oil pressure in the system, check for any clogged pipe that is creating this pressure build up or check for a faulty pressure switch. Ensure all valves are open according to valve chart.
Cooking Oil Storage Tank Low Level	This indicates that there is not enough oil in the storage tank. Please refill to an appropriate level or check for a faulty switch.
Cooking Oil Storage Tank High Level	This indicates that there is too much oil in the storage tank. Please drain until appropriate level has been reached or check for a faulty switch.
Fryer Cooking Oil Low Level	This indicates that there is not enough oil in the fryer kettle. Check the storage tank to ensure oil supply. Check level controls for operation. Check oil storage tank valve for operation. Check air supply to valves.
Fryer Cooking Oil High Level	This indicates that there is too much oil in the fryer kettle. Check the storage tank valve operation. Check level controls for operation. Check air supply to valves.
Paddle Conveyor AC Drive Communication Fault	This indicates there was loss of communication between the Paddle conveyor AC Drive and the PLC. Make sure that the AC drive is powered ON, check communication cables and communication module.
Cooking Oil Pump OFF at High Temperature	This indicates that the cooking oil circulation pump has been turned off with the oil at over 200 F. Turn ON and run pump until cooking oil cools down to at least 200 F.
Fryer Back Left E-Stop Active	This indicates that the emergency stop, located on the back left of the fryer is active.
Fryer Front Left E-Stop Active	This indicates that the emergency stop, located on the front left of the fryer is active.
Fryer Front Right E-Stop Active	This indicates that the emergency stop, located on the front right of the fryer is active.
Fryer Back Right E-Stop Active	This indicates that the emergency stop, located on the back right of the fryer is active.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

ALARM DESCRIPTIONS (continued)

Possible Alarm Messages	Description/Action
Fryer Control Panel E-Stop Active	Emergency Stop button on Fryer Panel has been Pressed.
Infeed Conveyor AC Drive Fault	This indicates a fault condition in the AC inverter located inside the fryer electrical panel. For a description of fault codes refer to the AB PowerFlex AC Drive manual supplied with the machine.
Infeed Conveyor AC Drive Communication Fault	This indicates there was loss of communication between the Infeed conveyor AC Drive and the PLC. Make sure that the AC drive is powered ON, check communication cables and communication module.
Discharge Conveyor AC Drive Communication Fault	This indicates there was loss of communication between the Discharge conveyor AC Drive and the PLC. Make sure that the AC drive is powered ON, check communication cables and communication module.
Fryer Air Pressure Fault	This indicates a low air pressure condition at the fryer. Verify the air pressure supply, check the air connections and any shut off valves. Verify the correct air pressure before starting the system.
Aspiration Conveyor Front Right E-Stop Active	This indicates that the emergency stop, located on the front right of the aspiration conveyor is active.
Aspiration Conveyor Front Left E-Stop Active	This indicates that the emergency stop, located on the front left of the aspiration conveyor is active.
Aspiration Conveyor AC Drive Communication Fault	This indicates there was loss of communication between the Aspiration Conveyor AC Drive and the PLC. Make sure that the AC drive is powered ON, check communication cables and communication module.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

SAFETY DURING DRAINING AND CLEANING OF FRYERS

Introduction

This information sheet advises on safety during emptying and cleaning of fryers. It includes guidance on manual draining and cleaning, as well as fryers with automated or semi-automated filtering (using enclosed portable filtering units).

Automated and semi-automated filtering processes avoid the need for the operator to come into contact with hot oil, significantly reducing the risks of burns. This enables oil filtering to take place safely even while the oil is at normal cooking temperature. In any case, most automated or semi-automated systems require an oil temperature of at least 212 F for the filtering process to work effectively.

Manual emptying and filtering of fryers should only be carried out when the oil has been cooled to 100 F.

Key Messages

Burns from hot oil can be very serious. Oil takes 30 minutes to heat up but can take 6-7 hours to cool down again.

Never add oil to Fryer when water is not completely drained!!

Whichever type of fryer is used, it is essential that:

- ◆ Staff are trained in safe procedures for draining and cleaning.
- ◆ Staff are provided with suitable protective equipment, where required by the risk assessment, eye protection, heat-resistant gloves and aprons.
- ◆ The fryer must be well maintained and any attachments used must be suitable for their purpose, as recommended by the manufacturer.
- ◆ Any Oil Leaks Must be caught by a Metal Container (**NOT PLASTIC**).
- ◆ Any oil spillages must be cleaned up immediately, ensuring floor areas around equipment are completely clean and dry to avoid slip risks.

When to drain and Clean

For fire safety and economy, cooking oil fryers must be switched off when unattended. It is best practice for manual oil filtering and cleaning to be carried out as a last task of the day rather than first task of the next day.

Hazards

Hazards connected with draining and cleaning fryers include:

- ◆ Fire

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

SAFETY DURING DRAINING AND CLEANING OF FRYERS (Cont.)

- ◆ Burns from hot oil
- ◆ Contact with hot surfaces
- ◆ Fumes from boiling cleaning chemicals and the danger of the chemicals overflowing
- ◆ Eye injuries from splashes
- ◆ Slips from oil spillage
- ◆ Strains and sprains from lifting and moving containers of oil.

If a 24-hour service is offered and the appliance is required continuously, there are two safe options:

- ◆ Use an automated filtering system or semi-automated portable filtering unit that removes the hot oil direct from the fryer, filters the oil and holds it safely

Automated and semi-automated filtering

Automated filtering systems

An automated system consists of an inbuilt oil filtration system. The oil is drained into an enclosed reservoir and an electric pump circulates it through a filter system and internal pipe work back into the fryer. Since this process is enclosed within the equipment, the operator does not come into contact with hot oil, greatly reducing any risk.

Portable oil Filtering Units (Semi-Automated)

These units are not part of the fryer but sit alongside it. The operator attaches an extension pipe to the fryer and the hot oil is drained into an enclosed container within the portable unit. The oil is then filtered as above and returned to the fryer.

If you have a fryer with an automated oil draining system or a portable oil filtering unit, you should refer to the manufacturer's guidelines for draining/filtering temperatures and safe operational requirements.

These, together with your own risk assessment, will determine the need for suitable protective equipment. If there is still a risk from contact with hot surfaces or oil splashing, you may require eye protection, a protective apron and/or heat-resistant gloves/gauntlets.

Manual Oil Filtering

This involves the operator draining the oil from the fryer, through a filter, into a oil storage tank.

Rules for draining oil safely and in the correct sequence can be based on the following guidelines:

- ◆ Allow the oil to cool, ideally for at least six hours, and check the temperature using a suitable probe thermometer before draining. Do not drain if the temperature is above 100 F.
- ◆ Follow any manufacturers instructions and use the correct equipment (eg a detachable spout for the

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

SAFETY DURING DRAINING AND CLEANING OF FRYERS (Cont.)

type of fryer being emptied), making sure any equipment required is brought to the fryer before the process start.

- ◆ Depending on the type of fryer, oil will be drained by a drain valve.
- ◆ If the oil is too cold to drain easily, reheat it briefly and run the conveyor (for no more than one minute). Switch the appliance off and check the temperature (using a suitable probe thermometer) before emptying.
- ◆ Using a filter, run the oil into a suitable metal holding or heat-resistant tank.
- ◆ Make sure that the container is empty and big enough to take the volume of oil being drained at any time.
- ◆ When large volumes of oil are being drained, it is safer to drain off in smaller amounts. This avoids overfilling the container and will reduce the chance of spillages when it is moved. Smaller amounts will also be easier to carry.
- ◆ Place the container in a safe place where it cannot be contaminated with chemicals, water or foreign bodies. Place the container on top of a drip tray to avoid any floor contamination.
- ◆ Do not dispose of waste oil down the drain. Disposal must comply with environmental legislation.
- ◆ Clean up any spillages **Immediately**.
- ◆ Make sure floor areas around equipment are completely clean and dry to avoid slip risks.

Other Precautions

- ◆ Make sure the design of the drain-off prevents it being turned on accidentally:
- ◆ Mark the tap clearly that it should not be touched.
- ◆ Place warning signs near the tap.
- ◆ If possible, remove the tap handle when the fryer is switched on.

Cleaning Procedure

- ◆ Turn Off the fryer and then On/Off control for gas.
- ◆ Make sure suitable protective equipment is worn, including eye protection (if appropriate). Check that other activities will not be put at risk by the cleaning activity.
- ◆ Check that other activities will not be put at risk by the cleaning activity.
- ◆ Check that the oil has been thoroughly drained and there are no spillages that may cause slipping.
- ◆ Remove loose debris from the internal surfaces.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

SAFETY DURING EMPTYING AND CLEANING OF FRYERS (Cont.)

- ◆ Thoroughly wash all internal and external surfaces with suitable cleaning chemicals and check for any leaks.
- ◆ For stubborn residues, fill the fryer with your recommended cleaning agents and leave or simmer according to chemical manufacturer instructions.
- ◆ Do not leave the fryer unattended or allow it to boil as this may cause the fryer to cascade liquid on to the floor, causing additional scalding and slipping hazards.
- ◆ Drain the appliance and rinse thoroughly with clean water until PH neutral.
- ◆ Dry all internal surfaces and make sure there is no water left in the fryer.
- ◆ Check the drain valve is closed and working properly then refill and switch “On” as required.
- ◆ When refilling the fryer with oil, some help may be required if the oil container is too large or heavy for one member of staff. Where possible use smaller containers.
- ◆ Do not overfill the fryer. Follow the manufacturer’s guidelines.
- ◆ Clean up any spillages **Immediately**.
- ◆ Make sure floor areas around the equipment are completely clean and dry to avoid slip risks.

Training

This section may apply to all types of fryers:

- ◆ Only staff who have been trained in safe use of the cleaning chemicals and cleaning procedures for the fryer should be allowed to do this task.
- ◆ Staff should be trained in the reporting procedures if they find the equipment to be faulty.
- ◆ Staff should be made aware of the reasons for use of suitable protective equipment, gloves and eye protection.
- ◆ Risk assessments should be completed for hazardous chemicals and staff should be made aware of the correct procedures for the use cleaning chemicals. Safety data sheets should be available to staff.
- ◆ A short, written procedure can act as a reminder to staff for both draining and cleaning operations.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

OIL EXTRACTOR FUNCTION AND THEORY OF OPERATIONS

The oil extractor is simply a grease type baffle filter enclosed in the stack of the fryer. When the steam exhaust blower is turned on steam is drawn out of the hood and through the filter. Heavier grease particles accumulate on the baffles and coagulate to form drops of oil, which are gathered in a catch pan and drained from the fryer hood. This reduces the amount of oil discharged from the stack as particulates in the steam.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER KETTLE OIL LEVEL SWITCH SETUP

1. When setting up the kettle oil level it is important to remember that the oil volume will rise when it is heated to operating temperature. If setting up the oil level when the fluid is cold, it is mandatory to evaluate and adjust the level later when the oil is at operating (frying) temperature.
2. The instructions included in this document are meant as a way to set up the oil level detection tubes from a "start" point out to a fully adjusted level ready for product frying. If you are simply adjusting the oil level AFTER the initial setup and making minor changes, skip to step 18.
3. Remove the side cover and the heat shield / cover from the oil level sensor box. *(See Fig. A)*



Fig. A

4. Remove the front cover from the level sensor pneumatics enclosure under the kettle. *(See Fig. B)*

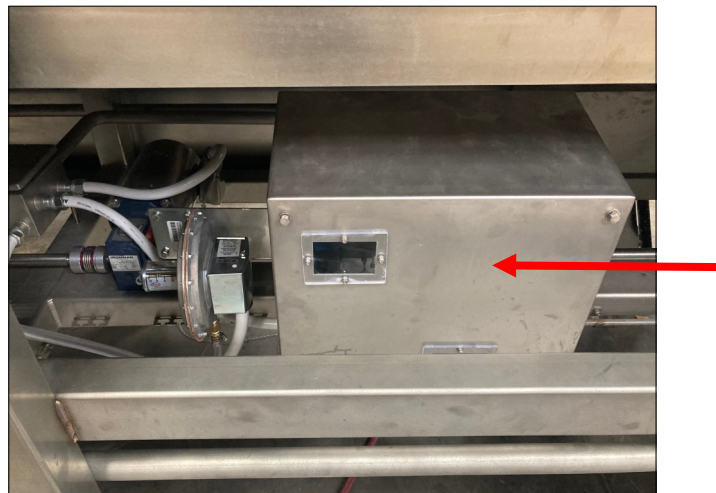


Fig. B

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER KETTLE OIL LEVEL SWITCH SETUP (Continued)

5. On the oil level sensor box, you will see two pipes leading into the oil. The high level float is at the rear of the oil level sensor box. The oil level bubbler is towards the front of the oil level sensor box. The bubbler is what actually monitors and sets the running oil level. *(See Fig. C and D)*



Fig. C

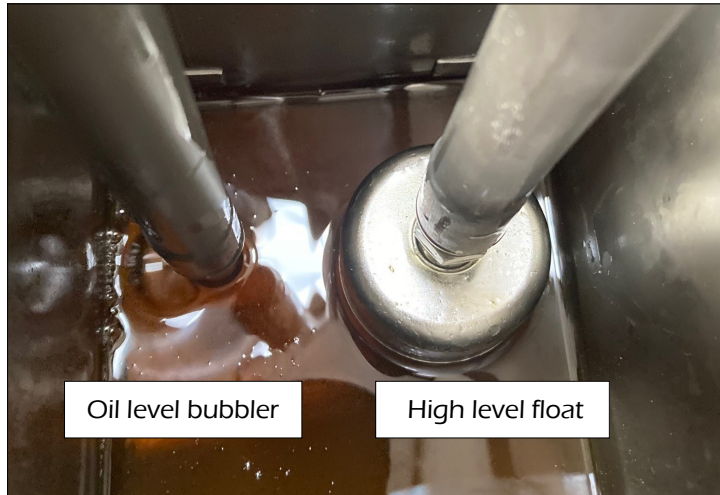


Fig. D

6. Both of the level probe pipes are secured with split collars mounted to the oil level box. To adjust either of them, use an Allen wrench to slightly loosen the collar screws for the probe you wish to adjust then carefully raise or lower the probe to the desired height. *(See Fig. E)*

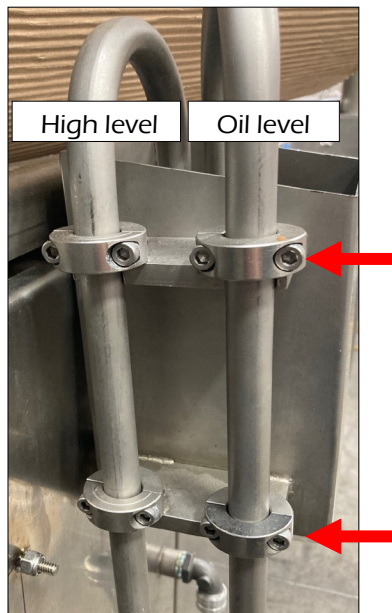


Fig. E

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER KETTLE OIL LEVEL SWITCH SETUP (Continued)

The next two steps are to setup a “Start” point for the oil operating level probe and the oil high level float. These are preliminary measurements which may change when the oil is at operating temperature.

7. Set the oil level probe to 5-1/2” from **TIP** of the **PIPE** to the top of the front side of the probe tank.
(See Fig. F and G)



Fig. F

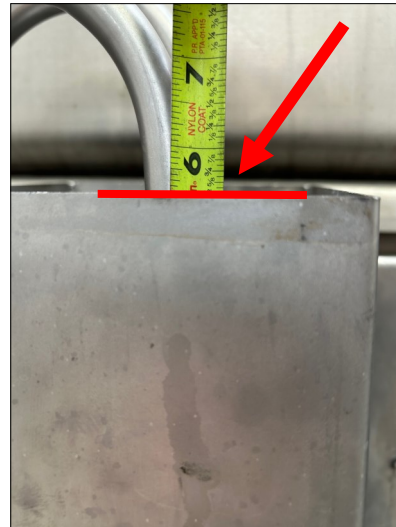


Fig. G

8. Set the oil High level float to 4-3/4” from **TOP** of the **FLOAT** to the top of the rear side of the probe tank.
(See Fig. H and I)



Fig. H



Fig. I

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER KETTLE OIL LEVEL SWITCH SETUP (continued)

9. In the Fryer Kettle oil level pneumatics enclosure, there are two adjustments to be made. The AIR FLOW SENSOR is on the top, right. The AIR PRESSURE REGULATOR is on the lower, left.
10. On the AIR PRESSURE REGULATOR, using the rotary knob on top of the regulator, adjust the AIR PRESSURE REGULATOR to read .500. (See Fig. J)



Fig. J

11. On the AIR FLOW SENSOR, using the rotary knob, Adjust the AIR FLOW to read 0.355. This adjustment sets the air flow rate out of the level probe. Air bubbles will be seen coming from the tube, this is normal, the sensor uses air backpressure to sense the oil level. When hot oil is introduced into the kettle this value may change, simply re-adjust this to read .355 (See Fig. K)



Fig. K

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER KETTLE OIL LEVEL SWITCH SETUP (continued)

12. Using a screwdriver, turn the slotted adjustment screw at the end of air pressure diaphragm switch. Turn the screw until the disk in the switch is lined up with .10. (See Fig. L)

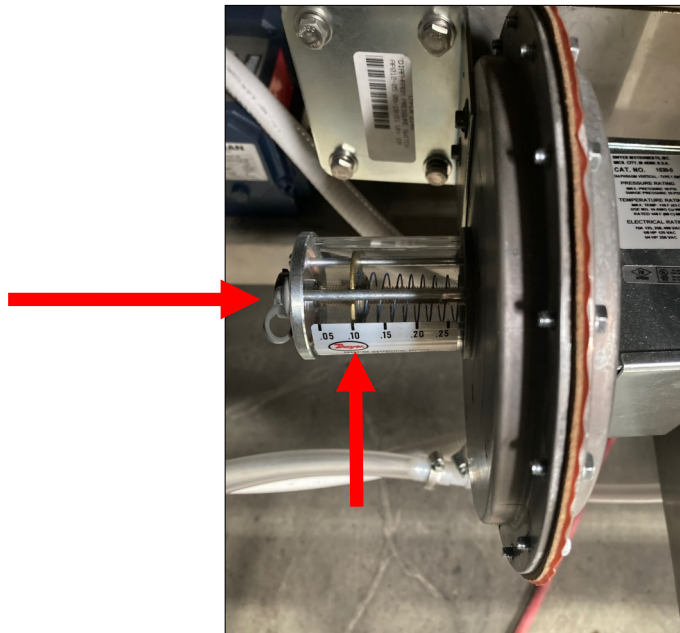


Fig. L

13. Fill the system with water or Oil. From the Fryer Control screen, turn on the Oil “Auto Fill” function. Wait for the fluid level to rise to the level probe then turn on the oil circulating pump. Listen for Cavitation or air in the pump. Stop the pump for 2 minutes to let the system “Burp” the air out. Restart the pump after the two minute pause.
14. From the Fryer Control screen, start the paddles, the submerger belt and the discharge belt.
15. Turn on the Burners (Boilers), If starting up from a sanitation or water is being used, set the oil temperature to 210 degrees Fahrenheit. If starting from a wash, allow the oil to “boil out” the remaining water for 30 minutes. If no water is present or are starting up from a previous oil production set the oil temperature to operating range (360).
16. When the cooking oil gets to operating temperature (360) it will expand and possibly go high level, this will cause the high level float to pump the excess oil out and send it to the storage tank. As the fryer runs and oil is consumed the oil level will balance out. It normally takes about 15 to 20 minutes to level out. At this time check the oil level, the level should be **OVER** the Submerger belt by about **1/4”**. Adjust the level probe if needed to achieve this setting. When the level probe is adjusted, raise or lower the high level float as well so that it is about 1/2” above the operating level of the cooking oil.
17. The following steps are a start to finish quick setup guide to set the oil level and high level float position.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER KETTLE OIL LEVEL SWITCH SETUP (continued)

17. Raise the high level float so that the float is not touching oil. We want the float at the bottom of the stem, above the oil level. We don't want the high level float to detect the oil level at this time.
18. Do not make large, quick adjustments with the oil level bubbler. Make the adjustment then wait for the results, it may take several minutes for the oil level to rise because it is metered in with very small volumes from the oil storage tank. The actual switching time can take 4 to 10 seconds to build backpressure or release the backpressure in the diaphragm switch.
19. **The oil level is to be set 1/4" above the chain mesh of the submerger belt.**
20. If the oil is at the desired level, Adjust the oil level bubbler so that the tip just contacts the surface of the oil, just enough to make a bubble escape the oil. *Do not fully submerge the entire probe, just the tip*. We want just enough penetration to make the low pressure air cause a backpressure in the oil level diaphragm switch. (See Fig. M)
21. If the oil level is lower than desired, adjust the tip of the probe to the approximate level and wait for the auto fill to continue filling the kettle. Be patient, the oil feed is very slow so not to overwhelm the heat exchanger and cause a low oil temperature situation. When the oil level is satisfactory, adjust the probe so that the tip barely contacts the oil. We want just enough penetration to make the low pressure air cause a backpressure in the oil level diaphragm switch. (See Fig. M)
21. If the oil level is higher than desired, lower the high level float so that it contacts the oil and turns on the drain pump to send some of the kettle oil back into the oil storage tank. When the oil level is satisfactory, raise the high level probe above the oil level. Now adjust the bubbler probe so that it is just penetrating the oil with the tip. (See Fig. M)
22. After the adjustments are made to the oil level bubbler, its time to adjust the high level float. Adjust the float so that the float contacts the oil and starts to float off the end of the stem. The high level float is not meant as a level control for normal operation, it is a secondary switch in case too much oil is added or the bubbler level control is not functioning. It is an emergency backup to the oil level bubbler. The drain pump should not be pumping oil into the storage tank while normally running.

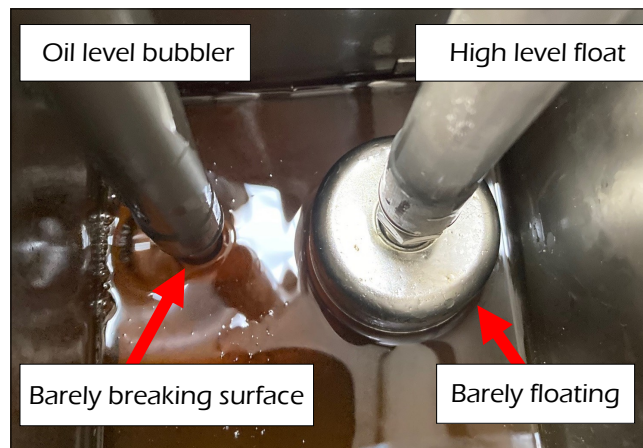


Fig. M

Due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

FRYER OIL FLOW

- * **Note:** Make sure oil is at proper level and at operating temperature. DO NOT close the valve completely while pump is running.
- 1. One critical adjustment on the fryer is the oil flow into the fryer bed. The flow should not be turbulent as the oil exits the manifold. Excess turbulence causes oxygenation of the oil.
- 2. Note: Oil flow will change with temperature oil is thinner at high temperature and more likely to have powerful turbulent flow.
- 3. To adjust the oil flow start the oil pump and double check oil level once oil is flowing and at temperature, slowly close the oil input Gate valve on the oil supply pipe. By turning the wheel counter clockwise. The threaded rod will travel down. *See Fig. A and B*
- 4. Close the valve until the oil flow looks “flat” without surface agitation and bubbles.

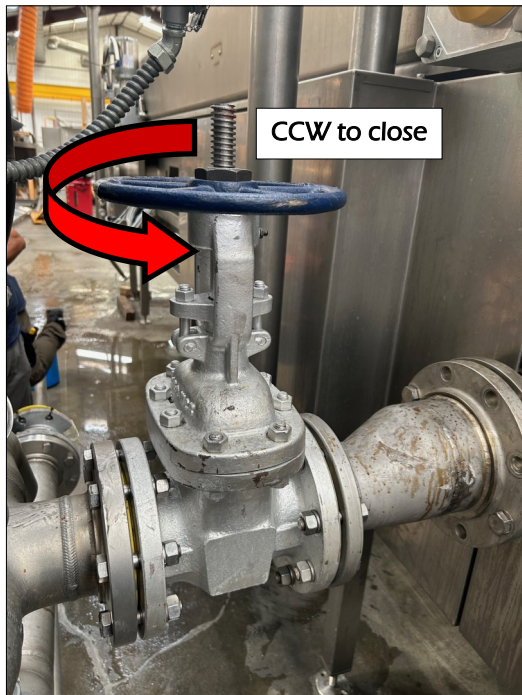


Fig. A

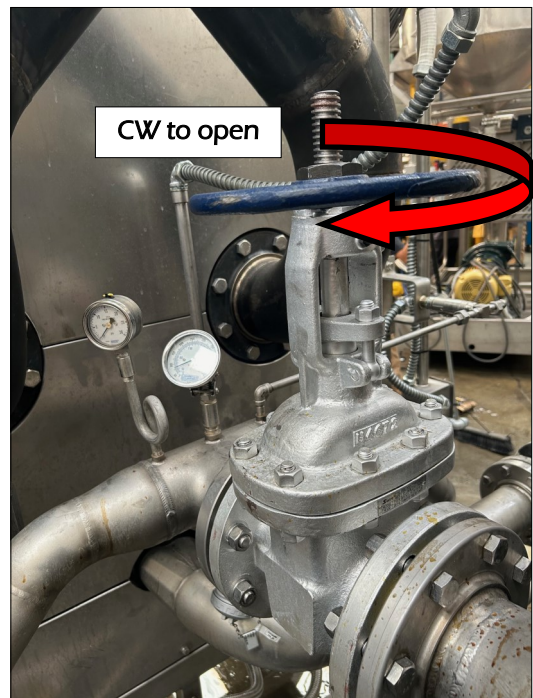


Fig. B